

An experimental syntax approach to British English *do* ellipsis

James Griffiths
University of Tübingen
james.griffiths@uni-tuebingen.de

Abstract

Researchers have reported that British English *do* (also called ‘propredicate *do*’) is permitted in only a subset of the syntactic configurations that permit standard VP / post-auxiliary ellipsis in English. Problematically, the reliability of the acceptability judgment data underlying all prior generative research on British English *do* ellipsis is not guaranteed. This situation transpires because these data were collected informally, which is not the most suitable method for collecting data on elliptical phenomena. In this paper, I report the results of eight judgment experiments conducted using the experimental syntax method, which **can** return reliable results on elliptical phenomena. These results show that, while the distribution of British English *do* ellipsis is indeed more restricted than standard post-auxiliary ellipsis, the *do* ellipsis site is permeable to all forms of syntactic movement from within it, contrary to reports in the literature. The results obtained are compatible with a simple syntactic analysis that states that *do* cannot select for elliptical VPs headed by a verb that can move to T. Eliding a VP in *do* ellipsis therefore does not render that VP opaque to further syntactic operations, which weakens the empirical support for proposals that treat ellipsis sites as phase-like objects.

Keywords: British English *do*, ellipsis, English dialects, experimental syntax, phases, propredicate *do*

1. Introduction

Speakers of British English can optionally place a non-finite version of *do* immediately before the site of post-auxiliary ellipsis (1). This is British English *do* ellipsis (also called ‘propredicate *do*’; see Joos 1964 for an early discussion), henceforth BDE.¹

- (1) a. I didn’t know the answer, but John may do. (Pullum & Wilson 1977:761)
b. I haven’t seen him, but John may have done. (ibid.)
c. John didn’t call her today, but he did do yesterday. (Chalcraft 2006:7)

Since the early 2010s, researchers have reported that BDE appears in only a subset of the syntactic configurations that permit standard VP ellipsis / post-auxiliary ellipsis (henceforth PAE). This diverging distribution has been used to support a variety of far-reaching theoretical claims, including that syntactic movement from within an ellipsis site is prohibited once ellipsis has been syntactically licensed (Aelbrecht 2010, Baltin 2012), that English PAE should be viewed as an instance of phasal spell-out (Bošković 2014, Lewis 2024), and that certain types of phrases but not others undergo syntactic reconstruction to the specifiers of phasal heads (Thoms & Sailor 2018).

Putting aside Chalcraft’s (2006) and Oger’s (2019) corpus-based analyses, all generative research on BDE to date has relied exclusively on acceptability judgment data collected informally from a handful of (presumably different) British English speaking consultants. As is well known (see Featherston 2020 and references in there), informal methods of data collection typically do not include safeguards for ensuring that consultants judge acceptability in absolute terms or for ensuring that relative judgments—that is to say, judgments about acceptability contrasts within minimal pairs of test sentences—are suitably controlled throughout a consultation session. Even if suitable controls **are** in place, research articles seldom make explicit the basis of comparison for relative judgments. Sprouse & Almeida (2012, 2017, 2018) and Sprouse et al. (2013) have shown that this methodological imprecision is unproblematic for most generative syntax research, as modulating syntactic factors typically yields large effect sizes. For instance, whether the asterisk accompanying (2b) represents a judgment about (2b)’s absolute acceptability or its acceptability relative to (2a) (or indeed any other grammatical English sentence) is unimportant, as the large negative effect on the acceptability of modulating basic word order will yield the same ‘asterisk’ judgment regardless of what benchmark the native speaker makes her judgment against.

- (2) a. Sally eats muesli every morning.
b. * Sally muesli every morning eats.

¹ Although British English *do* (henceforth *BD*) is also attested in media sources from Australia, Canada, New Zealand, South Africa, and the United States (Chalcraft 2006, Oger 2019), its productivity and prevalence in these varieties is not well known (about the Australian situation, Trudgill says in 1987 only that “the Australians are fairly happy with it”; cited in Kato & Butters 1987:213). In this paper, I uphold the naïve view that this phenomenon is exclusively British, purely to maintain terminological consistency across the literature.

Modulating the factors discussed in the BDE literature yields small effect sizes, however. In other words, the difference in acceptability between a putatively good and a putatively bad BDE configuration is much smaller than the difference between a sentence that obeys the basic word order rules of English and one that does not. The British English speaking reader can confirm this for herself by conducting an introspective within and between group comparison of the examples in (2) and (3). When dealing with small effect sizes, standards of comparisons become relevant again: (3b) might be unacceptable relative to (3a) but is certainly acceptable relative to (2b). Unfortunately, the standard of comparison for a particular judgment is never reported in the BDE literature. This can lead to situations in which the same BDE configuration is reported by one researcher as acceptable and by another as unacceptable. For concrete examples, compare (4a) and (4b), which both involve British English *do* in a pseudogapping configuration, and (5a) and (5b), which both involve topicalization from within the BDE site. The inherent ambiguity of ratings associated with phenomena with small effect sizes can also lead to situations in which researchers report as unacceptable sentences that are attested in spoken corpora or in print—this is analogous to a type I error. For instance, Lewis (2024) reports that ‘double *do*’ BDE configurations are unacceptable (6a), yet examples are readily found in corpora (Chalcraft 2006, Thompson 2014:106), in standard reference grammars (e.g., Quirk et al. 1985:875, Huddleston & Pullum 2002:101), and in online dialect archives (6b–c).

- (3) Although I don't know which book Fred will read, ...
 a. ... I do know which book Tom will.
 b. * ... I do know which book Tom will do. (Baltin 2007:31, his judgment)
- (4) a. Chomsky revolutionized English phonology, as earlier he had done English syntax.²
 (Pullum & Wilson 1977:762)
 b. * Although he wouldn't visit Martha, he would do Sally. (Baltin 2012:391)
- (5) a. * Hazelnuts I like, peanuts I don't do. (Haddican 2007:541)
 b. Hazelnuts I won't eat, peanuts I might do. (Abels 2012:31)
- (6) a. * Tom wrote a paper and Emma did do too. (Lewis 2024:203)
 b. We sat down next to one another and eventually I managed to pluck up the courage to ask him out for a drink with me. He did do—he said yes. www.dialectsarchive.com/england-14
 c. So when I was seventeen and they asked me if I could look after the shop, I did do. www.readingsheffield.co.uk/readers-stories-2/dorothy-norbury

Of course, the differing judgments attributed to structurally identical BDE configurations in the literature might have arisen from a different source, namely idiolectal variation. For instance, for (5a) versus (5b), Haddican's consultants may have each possessed a grammar in which mixing British English *do* with direct object topicalization is ungrammatical, whereas the opposite may have obtained for Abels' consultants. Unfortunately, the imprecision of the informal data-collection methodology makes it impossible to determine whether, for any specific disparity in judgments observed in the literature, the divergence is due to having no standard of comparison for judgments or instead reflects genuine idiolectal variation.

To state the issue in different terms: the informally collected data in (2) are informative because one can justifiably assume that the reported judgment is reliable (insofar as it would be replicated in a larger sample of speakers) and that the difference in acceptability reflects a change in grammaticality. These assumptions are warranted because of the large effect size involved. However, when the effect size of a modulation is small, as with BDE comparisons, holding these assumptions about informally collected judgment data is no longer justified. Such data are therefore underinformative, as one cannot know if a

² Kato & Butters (1987) claim that American speakers are particularly comfortable with the presence of non-finite *do* in *as*- and *which*-appositive clauses; see the attested examples in (i) and (ii) (see also Oger 2019). If so, there might be two non-finite *dos* in English: one which is present in all varieties and is licensed only in predicate-omitted appositives, and another which is exclusive to British English (and perhaps other commonwealth varieties) and is licensed in all types of predicate-omitted clauses. If this scenario obtains, and if these two *dos* display distinct formal properties, then (4a) and (4b) are not directly comparable, as they involve two distinct *dos*. But this is all speculative — determining the properties of a separate, English-wide ‘appositive-*do*’ (if such a phenomenon even exists) is far beyond this paper's remit. For attested examples of non-finite *do* across various configurations — not only appositive relative clauses — in a dialect of American English centred in Utah, see Di Paolo 1993.

(i) Mrs. Thatcher went off on her own and made good her threat to withdraw from UNESCO, as the United States had *done*. (Joseph Lelyveld, *The New York Times Weekly Review*, 22 December 1985) (Kato & Butters 1987:214)
 (ii) If this helps push the dollar down, as it has been *doing* since February, let the Fed's chairman, Paul Volcker, and his team rejoice, not desist. (Paul A. Samuelson, *The New York Times Weekly Review*, 15 September 1985) (ibid.)

judgment is reliable or if a change in acceptability reflects a change in grammaticality, in a syntactic sense.

The problems identified above can be circumvented by switching to the experimental syntax methodology. This method sets explicit standards of comparison for relative judgments and, when employed in conjunction with standardized fillers that designate the same floor and ceiling for any judgment across experiments (e.g., Gerbrich et al. 2019), returns absolute judgments. It also delivers a quantitative measure of effect size, thus providing researchers with the possibility to set an explicit lower-bound for what counts as a syntactic effect (as opposed to e.g. a prosodic or pragmatic effect) on acceptability. Furthermore, using sufficiently large sample sizes allows for patterns of idiolectal variation to be easily identified and permits the inherent noise associated with individual speaker judgments to be averaged over and/or treated as a cluster variable in mixed effects statistical models.

In November 2023, I conducted eight acceptability judgment tasks on BDE that use experimental syntax methods. The purpose of these experiments was to ascertain the representativeness of various judgments reported in the literature and to determine whether a degradation in acceptability reflects a syntactic violation or a non-syntactic one. The latter determination is made by considering the influence of effect size on statistical power. These eight experiments were designed such that they can only determine with confidence whether, for some factor influencing on BDE's acceptability reported in the literature (call it α), the underlying cause of α can be attributed to a violation of syntactic selection and/or locality constraints (as several recent analyses of BDE suggest; see §2.1). What these experiments cannot do is identify the underlying non-syntactic source of α 's effect. This situation arises because the experiments utilize designs and sample sizes known to have sufficient statistical power for detecting violations of syntactic locality conditions in elliptical configurations (see Cortés Rodríguez 2022 for experimental results on island and clausemateness violations in multiple sluicing; see Schiele 2025 for experimental results on left-branch extraction violations in German fragments) but which might not be sufficiently powered to detect smaller, non-syntactic effects.³

In this paper, I report the results of these experiments and then utilize them to develop a new syntactic analysis of BDE. The paper starts in earnest in §2.1 with a brief sketch of four recent analyses of BDE. This sketch performs several functions. Firstly, it draws our attention to the subclass of restrictions on BDE that have been the primary focus of prior research, namely restrictions on combining BDE with phrasal movement, where the moving phrase could be viewed—depending on one's background assumptions—as originating within the BDE site (see (3b), (4b), and (5a) for purported examples). For brevity's sake, I henceforth refer to these restrictions simply as *movement restrictions*. Secondly, the sketch serves to demonstrate the diversity of empirical and conceptual claims about the movement restrictions on BDE present in the literature and, by doing this, it—thirdly—emphasizes the need for replicating the relevant reported judgments using experimental methods. The results of six experiments that focus on the movement restrictions on BDE are then presented in §2.2 to §2.6. The main finding from these experiments is that, at least from a syntactic perspective, there are **no** movement restrictions on BDE.

As explained in §4, in which I return to critically assess the four analyses of BDE introduced in §2.1, these experimental results invalidate all prior analyses, including the influential idea that BDE of a phrase XP causes XP to become a barrier for syntactic movement across it (Aelbrecht 2010, Baltin 2012, Lewis 2024). Considering that explaining the alleged barrier-inducing properties of BDE was one of the original motivations behind the *derivational* approach to ellipsis (e.g., Aelbrecht 2010, Bošković 2014, Sailor 2018, Manetta 2020; see Güneş & Lipták 2022 for an overview), the results reported in this paper provide new motivation for reassessing the validity of this approach to ellipsis more generally. In other words, these results cast some doubt on the idea that licensing an XP for ellipsis causes the content of XP to be unavailable as an input for syntactic operations at future points in syntactic derivational-time.

BDE is not entirely free from syntactic constraints, however—for instance, BD is incompatible with *be*-passive and copular phrase ellipsis sites. In §3, I present new data to support the novel generalization that BD is only compatible with an elliptical phrase which tops the thematic verbal domain and whose head is a non-raising verb, that is to say, a verb that can never, in any English syntactic configuration, move to T. In §5, I present a simple syntactic analysis of BDE that captures this generalization, which treats British English *do* as a semantically vacuous light verb that selects only for non-raising verbs. In §6, I discuss outstanding issues and, in §7, I provide concluding remarks.

A final prefatory remark: The question of whether syntacticians working on small effect size phenomena should abandon informal data-collection methods altogether or are justified in employing such methods under certain circumstances will be sidestepped here. By adopting the pragmatic position

³ To be clear: I assume for the purposes of this study that, *ceteris paribus*, a violation of syntactic selection or syntactic locality conditions yields a larger negative effect on acceptability than atypical prosodic word or phrase formation does, and that the experiments reported in §§2-3 are designed such that statistical significance is the minimal requirement for treating a negative modulation as caused by a violation of syntactic selection or syntactic locality conditions. I thank an anonymous reviewer for pushing for further clarity on this point.

that informal methods *will* continue to be used, I aim with this paper to highlight the importance of *replication*, that is to say, of formally (dis)confirming informally collected judgments reported in the literature. Indeed, I myself will report new informally collected judgment data on BDE in §3. My hope is that an attempt to replicate these judgments will be undertaken in the near future, in order to confirm their reliability.

2. Six acceptability judgment experiments on the movement restrictions on BDE

2.1. Motivating the movement experiments: A sketch of four recent analyses of BDE

Since the early 2010s, researchers have reported that sentences containing both BDE and some types of phrasal movement that could be analyzed as originating within the ellipsis site are judged as unacceptable. If true, then BDE contrasts with PAE, which displays no such movement restrictions, once extraneous constraints are satisfied.⁴ However, the assumptions adopted and/or claims made about precisely which types of movement BDE is incompatible with differ across studies, resulting in a situation in which researchers have developed disparate analyses in order to capture distinct datasets. I contend that these studies differ in their empirical claims because they rely on acceptability judgment data that were collected informally, which is an unsuitable method for collecting data on phenomena with small effect sizes such as BDE, for the reasons already outlined in §1.

In order to provide an impression about the extent to which the empirical and conceptual claims about the movement restrictions on BDE differ across studies, I will now briefly outline four reasonably recent analyses of BDE, namely Aelbrecht 2010 / Baltin 2012,⁵ Thompson 2014, Thoms & Sailor 2018, and Lewis 2024. By doing this, I also provide the necessary background and motivation for the six movement-related experiments reported in the following subsections of §2.

Several researchers have interpreted the movement restrictions on BDE as support for the *derivational* approach to ellipsis, according to which elided phrases are treated as (displaying the same characteristics as) spelled-out phases and/or phase complements. According to this view, a phrase XP becomes opaque for further syntactic operations at the moment in derivational-time at which XP is syntactically licensed for ellipsis (Aelbrecht 2010, Baltin 2012, Bošković 2014, Lewis 2024). Analyses that adopt a derivational approach appeal to the derivational-timing of ellipsis licensing to explain why BDE exhibits movement restrictions whereas PAE does not. Each analysis claims that phrases of all types can escape XP before XP is syntactically licensed for ellipsis in PAE configurations, whereas some types of phrases fail to escape XP before XP is syntactically licensed in BDE configurations.

These analyses differ in their technical details. For instance, Aelbrecht (2010) and Baltin (2012) claim that BDE involves deletion of a smaller-sized phrase than PAE and that BDE is syntactically licensed by a c-commanding head H sooner – derivationally speaking – than PAE. These two differences conspire to ensure that, although all forms of phrasal movement can escape the PAE site before XP becomes syntactically frozen (see (7a); where chevrons indicate spell-out), only phrasal movement that can pit-stop in Spec,HP (namely, the movement that derives unaccusative clauses) can escape the BDE site before XP becomes frozen (compare (7b) and (7c)). The Aelbrecht-Baltin analysis therefore predicts that BDE is incompatible with all non-agentive phrasal movement aside from unaccusative-raising, and it makes the empirical claim that the movement restrictions on BDE indeed distribute as such, meaning that, as far as Aelbrecht/Baltin are concerned, their analysis correctly captures the movement restrictions on BDE.

- | | | |
|-----|--|-----------------|
| (7) | a. YP escapes XP before XP is spelled-out
[_{HP} H _{LICENSOR} ... [YP ₁ ... <[_{XP} ... t ₁ ...]>]] | <i>PAE</i> |
| | b. YP escapes XP via Spec,HP just before XP is spelled-out
[_{HP} YP ₁ [_{H'} H _{LICENSOR} <[_{XP} ... t ₁ ...]>]] | <i>good BDE</i> |
| | c. YP cannot escape XP before XP is spelled-out
[_{HP} H _{LICENSOR} <[_{XP} ... YP ...]>] | <i>bad BDE</i> |

⁴ A'-movement of XP from the PAE site is permitted only in presence of a focus-marked item F, where F must c-command the PAE site (or be able to reconstruct into the PAE site) and is c-commanded by the final landing site of XP (Griffiths 2019). This condition is satisfied in (i) and (ii), but not in (iii) (where grey shading denotes ellipsis, and small caps denotes focus). See Griffiths 2019 and Stockwell 2020 for recent analyses.

(i) I know who JOHN will kiss and also who₁ MARY will [kiss t₁].

(ii) I know which GIRL he kissed, but not [which BOY]₁ he did [kiss t₁].

(iii) * SUE doesn't know who he kissed, and MARY doesn't know who₁ he did [kiss t₁], either.

(Griffiths 2019:588)

⁵ The analyses offered in Aelbrecht 2010 and Baltin 2012 are very similar, differing only in details irrelevant for current purposes. Therefore, I treat them as one analysis in this paper.

Lewis (2024), on the other hand, claims that BDE always targets a full phase. BDE therefore contrasts with PAE, which is viewed by her as targeting either full phases *or* phasal complements (Bošković 2014). Following the standard phase-theory logic, ellipsis is syntactically licensed only when the next-highest phasal head enters the syntactic derivation. In this system, phrasal movement from the BDE site is possible if it occurs before the next-highest phasal head is first-merged, which is true of some types of phasal movement (e.g., topicalization, quantifier raising, subject raising) but not others (e.g., object *wh*-movement, long distance subject *wh*-movement). Because PAE can target a phase complement YP, all phrasal movement from YP can escape via the local phase-edge and is therefore never trapped in YP (8). Lewis makes the empirical claim that the movement restrictions on BDE distribute exactly as her analysis predicts, meaning that, as far as she is concerned, her analysis correctly captures the movement restrictions on BDE.

- (8) a. YP escapes XP through the local phase-edge before XP is spelled-out PAE
 [WP=PHASE W ... [ZP=PHASE YP₁ <[XP=PHASE-COMPLEMENT ... t₁ ...]>]], *or*
 [WP=PHASE W ... [YP₁ ... [ZP=PHASE t₁ <[XP=PHASE-COMPLEMENT ... t₁ ...]>]]]
 b. YP escapes ZP before ZP is spelled-out good BDE
 [WP=PHASE W ... [YP₁ ... <[ZP=PHASE t₁ [XP ... t₁ ...]>]]]
 c. YP cannot escape ZP before ZP is spelled-out bad BDE
 [WP=PHASE W ... <[ZP=PHASE YP₁ [XP ... t₁ ...]>]]

Not all recent analyses of BDE advocate for a derivational ellipsis approach. For instance, Thompson (2014) claims that BDE is compatible only with *covert* phrasal movement (i.e., quantifier raising and operator movement; see Table 1), and she captures this by proposing that British English *do* selects for a null verbal proform in the narrow syntax, which is replaced by a copy of the corresponding phrase in the antecedent at LF (so-called *LF-copying*; see Chung et al. 1995), from which LF-movement can proceed (9).

- (9) A man will read every book, and a woman will do, too. (Thompson 2014:244)
 a. ... [a woman]₁ will [_{VP} t₁ do Δ_{VP}] narrow syntax
 b. ... [a woman]₁ will [_{VP} t₁ do <[_{VP} read every book]>] at LF; <> = the LF-copied phrase
 c. ... [every book]₂ [[a woman]₁ will [_{VP} t₁ do <[_{VP} read t₂>]]] at LF, after QR

Another analysis that avoids recourse to derivational ellipsis comes from Thoms & Sailor (2018). They claim that BDE is incompatible only with phrasal movement operations that can undergo syntactic reconstruction (see Table 1), and which therefore must leave intermediate copies of movement (as opposed to null operators) in Spec,vP as they move successive-cyclically from within the BDE site. To capture this, they offer an analysis that appeals to BD's status as an enclitic (Haddican 2007). They suggest that there are no syntactic restrictions on extraction from the BDE site but that movement from within it sometimes blocks BD from cliticizing leftward, which yields prosodic ill-formedness. Blocking occurs only when a full-fledged syntactic copy linearly intervenes between BD and its potential prosodic host. If the intervenor is the copy of a null operator, then no blocking occurs, as null operators are entirely invisible to PF (10).

- (10) a. * [YP ... WP_{HOST} [_{VP} {YP} -do <[XP ... {YP} ...]>]] {} = a copy of movement,
 b. [Op ... WP_{HOST} [_{VP} {Op} -do <[XP ... {Op} ...]>]] and <> = the ellipsis site

One observes that the four studies outlined above adopt different views on the basic nature of ellipsis (as akin to phasal spell-out, as LF-copying, or as a superficial non-pronunciation operation), and that these different views arise – at least partly – from attempting to capture disparate datasets related to BDE's compatibility with movement. (The varying empirical landscapes that these four analyses attempt to capture are summarized in Table 1. This table describes, for each type of movement *M*, whether a particular study reports and/or predicts *M* to be compatible with BDE.)⁶ Consequently, this sketch has demonstrated that insights obtained from studying the movement restrictions on BDE are likely to be highly informative for answering big-picture questions about the nature of ellipsis, and that determining

⁶ In Table 1, “✓” denotes that a study both reports acceptability for that movement-type and that its analysis predicts acceptability; “(✓)” denotes that a study does not provide an acceptability judgment for that movement-type, but its analysis predicts acceptability; and “[✓]” denotes that a paper reports acceptability for that movement-type but that its analysis predicts unacceptability. (The same system holds for “✗”, “(✗)” and “[✗]”, but mirrored.) “Unknown” denotes that more information is required before it can be determined whether the analysis predicts that the specified movement-type is compatible with BDE.

which of these views on ellipsis is most likely to be correct is contingent on obtaining reliable acceptability judgment data on the movement restrictions on BDE.

Table 1. Stated and/or predicted movement restrictions on BDE, according to four recent analyses

Type of movement		stated / predicted compatibility with BDE			
		Aelbrecht 2010, Baltin 2012	Thompson 2014	Thoms & Sailor 2018	Lewis 2024
overt A'-movement	<i>wh</i> -object	x	x	x	x
	long distance <i>wh</i> -subject	✓	✓	✓	x
	object topicalization	x	x	✓	✓
	raising-style relativization	(x)	(x)	✓	(unknown)
covert A'-movement	pseudogapping	x	x	(unknown)	(✓)
	operator movement	x	✓ ⁷	✓	(unknown)
	quantifier raising	x	✓	✓	✓
overt A-movement	subject raising	✓	✓	✓	✓
	unaccusatives	✓	(x)	✓	✓
	<i>be</i> -passivization	x	x	[x]	x
	<i>get</i> -passivization	(x)	x	✓	✓

In the following subsections, I report and interpret the results of six acceptability judgment experiments on the movement restrictions on BDE. These experiments – plus the two reported later in §3 – were created and conducted as part of the advanced undergraduate seminar *Data Collection in Linguistics*, which I taught in the winter semester of 2023-24 at the University of Tübingen. The experiments were designed, populated, and analyzed by 10 small groups of (under)graduate students, but supervised, conducted, and reanalyzed by me. Before my students and I designed the experiments, I held informal consultation sessions with 5 speakers of British English (all male, all in their 30s, from different parts of England). The informal judgments obtained from these sessions often served as an impetus for including particular factors in an experiment's design, as will become evident shortly.

2.2. Experimental procedure and inferential statistics

The experiments reported in this paper were 1–7 Likert scale acceptability judgment studies (1 = completely unnatural, 7 = completely natural) developed using *PsychoPy3* (Peirce et al. 2019). Each experiment consisted of a consent screen, 4 demographic screens, 5 instruction screens, a practice round, and a trial round. The experiment collected demographic information on gender, age (from prespecified ranges), and multilingual profile (if applicable), and asked the participant to confirm her status as a native speaker of British English. The practice round consisted of 7 standardized fillers and the trial round consisted of the test items and remaining fillers, presented in a random order. Participants that identified as non-British English speakers or who refused to provide consent were automatically ejected early and received no payment. All stimuli consisted of two parts, a *context* (presented in a smaller font size) and a *target* (larger font size). For the test items, the antecedent clause functioned as the context, and the elliptical clause as the target. Participants were instructed to judge only the target, but to take the context into consideration when making their judgments. All but one experiment used the same set of 28 fillers in the trial round. Of these 28 fillers, 20 were standardized fillers for English developed by Gebrich et al. (2019) (4 x each A to E level) and 8 were a balanced mixture of undisputedly acceptable and unacceptable dialogues involving ellipsis. The exceptional experiment, referred to as *Pseudogap* in §2.6, used half of these fillers (i.e., 10 standardized fillers and 4 elliptical fillers). The practice round consisted of 7 standardized fillers adapted from Sprouse et al. (2013)—one for each point on the 1–7 scale.

Gender and age-range were recorded to ensure that a representative sample was obtained for each experiment.⁸ Standardized fillers were employed in the practice and trial rounds to guard against scale bias and distortion and to provide standards of comparison for absolute judgments. The same filler items were used in (almost) every experiment to guarantee that, if needed, z-scored ratings—z-transformed using only scores from fillers, as opposed to scores from all experimental stimuli (Gebrich et al. 2019)—could be used to conduct viable cross-experiment comparisons.

How the test items were developed, arranged, and presented to participants depended on the experiment's design. Table 2 provides a summary. In all experiments, test items were distributed in a

⁷ Thompson reports that BDE is not permitted in *that*-relatives, which she assumes are generated by operator movement. Because her analysis predicts that BDE is compatible with configurations generated by covert phrasal movement, she concedes that she cannot explain why BDE is not permitted in *that*-relatives.

⁸ See the Supplementary Materials for evidence that representative samples were obtained for every experiment.

Latin square configuration over either 2, 4 or 8 lists (see Table 2 for details), with each participant assigned to only one list. In the 2x2 experiments, no list contained two test items from the same lexical set, whereas, in the 2x2x2 experiments, each list contained two test items from the same lexical set (but always for different conditions). All of the experimental items are listed in the Supplementary Materials.

Table 2. Summary of test item set-up, across all 8 experiments

Exp design	number of:				
	experiments	reps per condition	lexical sets	test items	lists
1x2	1	3	6	6	2
2x2	4	3	12	12	4
2x2	1	3	24	24	8
2x2x2	2	3	12	24	4

All eight experiments were unsupervised online studies, hosted by *pavlov.org*. Participants were recruited from *Prolific* and were screened for their location (UK and Ireland), the place of most time spent before turning 18 (UK and Ireland), primary language (English), first language (English), and approval rating on *Prolific* (95–100%). The experiments were conducted between 30 Nov and 07 Dec 2023. Participant remuneration was based on a £9 hourly rate. The 1x2 experiment took on average 6 minutes to complete, the 2x2 experiments took 9 minutes, and the 2x2x2 experiments took 12 minutes. No participant was involved in any study more than once. The sum of squared error (SSE) method was employed for identifying outliers, with the threshold set as 2 standard deviations above the mean SSE for filler judgments. Outliers were excluded from statistical analyses.

Cumulative Link Mixed Models (CLMMs) were fit to the raw scores using R's *ordinal* package (Christensen 2023). All CLMMs display a maximal fixed effect structure and the best fitting random effect structure, as determined manually through maximum likelihood testing (R's *anova* function). All CLMMs have *anova*-coded predictor variables. The model estimates, z values, and p values reported in §3 and §4 were generated using *ordinal*'s *summary* function. At the request of an anonymous reviewer, the results for all conditions across all experiments were checked for multimodal distributions, and none were observed.

2.3. *Wh*-object extraction and object topicalization

Recall that each of the exemplar analyses sketched in §2.1 assumed that BDE is incompatible with *wh*-object extraction (see Table 1). These analyses reflect the consensus position in the literature, which is that object *wh*-movement is permitted from within a PAE site but precluded from within a BDE site (Chalcraft 2006, Baltin 2006, 2007, 2012, Haddican 2007, Aelbrecht 2010, Thoms 2011, Thoms & Sailor 2018, Lewis 2024):

- (11) Although I don't know which book Fred will read, I do know [which book]₁ Tom will (*do) [read *t*₁].
(Baltin 2007:31)

Researchers have disagreed about whether object topicalization is compatible with BDE, however. The current stance is that object topicalization is permitted (Abels 2012, Thoms & Sailor 2018, Lewis 2024) (12a). As mentioned in §1 and implied §2.1, this view has superseded the stance held by Haddican (2007), Aelbrecht (2010) and Baltin (2012), who report and/or predict that object topicalization is incompatible with BDE (12b).

- (12) a. Hazelnuts I won't eat, peanuts I might do. (Abels 2012:31)
b. * Hazelnuts I like, peanuts I don't do. (Haddican 2007:541)

Absent from the literature are any reports on the compatibility of BDE with the topicalization and *wh*-movement of adjuncts, however. Generally speaking, BDE is compatible with both TP-adjoining and VP-adjoining adjuncts, as the examples in (13) show. This demonstrates that BDE can—and perhaps *must*—target a phrasal node that is properly contained in the phrasal node to which these adjuncts adjoin. Based on this, one is assured that A'-moving such adjuncts via topicalization or *wh*-movement involves movement from a base-position outside of the BDE site. The absence of reports on the compatibility of BDE with *wh*-adjunct movement and adjunct topicalization means that it has not yet been demonstrated that *wh*-object movement and object topicalization degrade acceptability (if indeed either does) because they move from within the BDE site, as opposed to degrading acceptability due to violating a more general yet currently unknown constraint that affects all non-subject *wh*-movement / topicalization in a clause containing BDE.

- (13) a. We don't often go out at weekends, but we do do *sometimes*. (Chalcraft 2006)
 b. Has anything like that ever happened to you? Yes, it does do *regularly*. (ibid.)
 c. I'm not working today, but I will be doing *tomorrow*. (ibid.)
 d. John said he would help, but he doesn't *usually* do. (ibid.)
 e. John didn't call her today, but he did do *yesterday*. (ibid.)
 f. Although he wouldn't bake a cake for Sally, he would do *for Mary*. (Baltin 2007:32)
 g. Although he wouldn't visit Sally on her birthday, he would do *on her anniversary*. (ibid.)

I presented my consultants with BDE configurations that involved topicalization and *wh*-movement of direct objects (extracted from within the BD ellipsis site) and adjuncts (extraction from outside of the BD ellipsis site), see (14). They reported that no difference in acceptability obtains between any case.

- (14) a. Jason's song I won't sing, but Ava's song I might do. *inside*
 b. With Sam I can't discuss the problem, but with Tom I can do. *outside*
 c. I know which shoes I can reserve for Steve, and also which ones I can't do. *inside*
 d. I know where Pete should watch TV, and also where Sue should do. *outside*

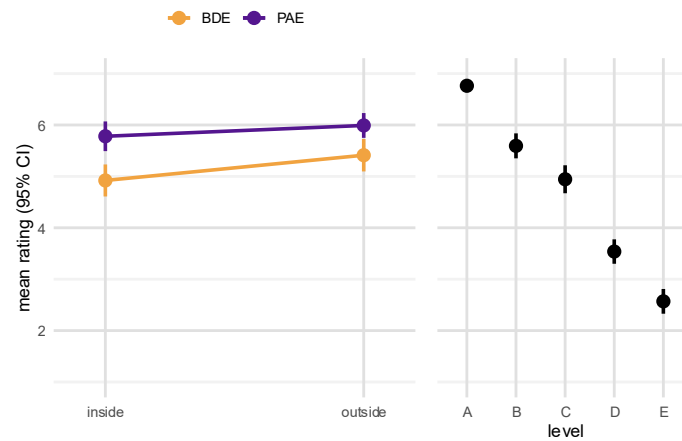
The Topicalization and *Wh*-object experiments aimed to determine whether my consultants' informal judgments regarding (14) aligned with those collected under experimental conditions from a more representative sample of the population. Both experiments had 2 two-levelled factors, namely ELLIPSIS (*BDE*, *PAE*) and POSITION (*inside*, *outside*), where POSITION refers to whether the tail of the A'-chain originates inside or outside of the BDE site. The experiments tested the possibility that topicalization / *wh*-movement from within the BDE site is less acceptable than topicalization / *wh*-movement from a non-subject position outside of the BDE site, relative to their PAE counterparts (as mentioned already in §2.1, it is undisputed that A'-extraction from within the PAE site is permitted, provided extraneous conditions are met; see Takahashi & Fox 2005, Merchant 2008b, Messick & Thoms 2016, Griffiths 2019). Sample test conditions have already been provided in (14) (to obtain the PAE conditions from the list in (14), one must simply omit *do* from each example). Note that the examples in (14) come from different lexical sets (within a lexical set, variation between items was kept to a minimum).⁹ Note also that the verb that immediately precedes the ellipsis site (in the case of PAE) or that immediately precedes *do* (in the case of BDE), which I henceforth call the 'introducer verb', was always a modal verb. Put differently, none of the test items in these two experiments used the *done* version of British English *do*. Furthermore, in order to keep things manageable, none of the items in any experiment fits the 'double *do*' template exemplified in (6b–c) or (12b).

Starting with the Topicalization experiment, 40 participants took part, with 2 identified as outliers. Judgments from 38 participants were therefore included in the analysis, meaning 114 observations per condition. The mean ratings for the test items and standard fillers are plotted in Figure 1.¹⁰

⁹ Small variations in word choice, sentence length, etc. are included across lexical sets to ensure that inferences from experimental results are robust. Such variations are factored into statistical analyses through modelling ITEM as a random effect. The position of contrastive focus – whether it is borne by the finite auxiliary verb or not – was also varied across lexical sets. This was not included as a fixed effect in any experiment because no prior study has identified this variable as affecting the licensing of BDE. An anonymous reviewer wondered whether this decision was correct (perhaps because Pullum & Miller (2013) have shown that the position of contrastive focus affects the licensing of exophoric PAE). To address the reviewer's concerns, in addition to fitting the model described below in the main text, I also fit a model that included FOCUS-PLACEMENT (focus on aux, focus on non-aux) as a fixed effect and found that, although focus on the auxiliary verb slightly increases acceptability for all the conditions, it has no specific effect on the licensing of BDE. I therefore view this null result as justification for not including FOCUS-PLACEMENT as a fixed effect variable in any other statistical model reported in this paper.

¹⁰ See §3 of the Supplementary Materials for the exact mean ratings (plus standard deviations and frequencies) for the test items and standardized fillers for all experiments.

Figure 1. Mean ratings for test and standardized filler items for the Topicalization experiment



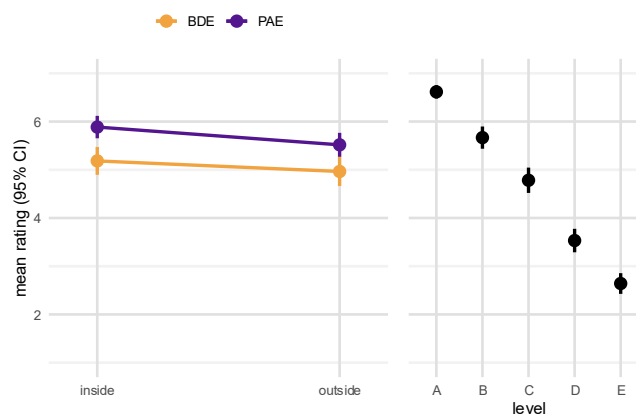
A summary of the fixed effect estimates for the best fitting CLMM for the Topicalization data is presented in Table 3. This summary shows significant main effects for ELLIPSIS and POSITION, but no significant interaction effect. The significant main effect returned for ELLIPSIS demonstrates that, generally speaking, PAE is judged as significantly more acceptable than BDE. I interpret this difference, which is replicated throughout my experiment series, as demonstrating merely that, when compared to PAE, BDE is treated by native speakers as a marked configuration, which is unsurprising when one considers that *do*'s presence is always optional and therefore always redundant. I suggest that this markedness is reflected in the lower acceptability ratings for BDE. Similarly, the significant main effect returned for POSITION is expected, as the linear distance between the filler and its gap is longer (and therefore harder to process, resulting in degraded acceptability) in the *inside* condition, regardless of ellipsis-type. Of more interest here is the interaction effect, whose insignificance strongly suggests that the degree of added degradation in acceptability yielded from topicalizing a direct object from within a BDE site (as opposed to topicalizing an adjunct from outside of the BDE site) is too small to be caused by violating a syntactic locality condition, in accordance with Abels 2012, Thoms & Sailor 2018, and Lewis 2024, but *contra* Haddican 2007, Aelbrecht 2010, and Baltin 2012.

Table 3. Summary of interaction model for the Topicalization experiment

score ~ position * ellipsis + (ellipsis item) + (1 subject)				
Contrasts	Estimate (β)	SE	z	p
PAE – BDE	1.41	0.32	4.23	<0.01
outside – inside	0.63	0.32	1.97	0.05
(PAE_outside – PAE_inside) – (BDE_outside – BDE_inside)	–0.73	0.63	–0.69	0.24

Turning to the *Wh*-object experiment, 39 participants took part, with 1 identified as an outlier. Judgments from 38 participants were therefore included in the analysis, meaning 114 observations per condition. The mean ratings for the test items and standard fillers are plotted in Figure 2.

Figure 2. Mean ratings for test and standardized filler items for the *Wh*-object experiment



A summary of the fixed effect estimates for the best fitting CLMM for the *Wh*-object data is presented in Table 4. This summary shows a significant main effect for ELLIPSIS and a marginal main effect for POSITION, but no significant interaction effect. Regarding the insignificance of the interaction effect, this result strongly suggests that the degree of added variation in acceptability yielded from *wh*-extracting a direct object from within a BD ellipsis site (as opposed to *wh*-extracting an adjunct from outside of the BD ellipsis site) is too small to be caused by violating a syntactic locality condition, *contra* Aelbrecht 2010, Baltin 2012, and Lewis 2024.

Table 4. Summary of interaction model for the *Wh*-object experiment

score ~ ellipsis * position + (1 item) + (ellipsis subject)				
Contrasts	Estimate (β)	SE	z	p
PAE – BDE	1.18	0.40	3.02	<0.01
outside – inside	0.62	0.34	1.81	0.07
(PAE_outside – PAE_inside) – (BDE_outside – BDE_inside)	–0.49	0.68	–0.63	0.53

Interestingly, Figure 2 reveals an unexpected preference for *wh*-extraction of direct objects as opposed to adjuncts in both PAE and BDE configurations. Precisely why this preference is observed is unknown to me. Several of the test items in the *outside* condition involved *wh*-pied-piping (e.g., ... *and also to whom I should've*): I tentatively suggest that participants may have perceived such configurations as marked—as *wh*-pied-piping could be construed as overly formal—thus introducing an extraneous and unwanted negative influence on acceptability into the *outside* condition.

2.4. *Wh*-subject extraction

Griffiths & Den Dikken (2022:125) report that BDE is incompatible with *wh*-subject movement:

- (15) a. A: Sue wouldn't kiss Peter last night.
 B: Well, who WOULD (*do)?
 b. If even Sue wouldn't kiss Peter, I don't know who WOULD (*do)!

However, Lewis' (2024) six British English consultants disagree, judging the examples in (15) as fully acceptable when *do* is present. Gary Thoms (pers. comm.) adopts the middle ground, judging the examples in (15) as degraded but improved to full acceptability when *have* is present as the introducer verb (16). My consultants agree with Thoms that the examples in (16) are more acceptable than their counterparts in (15).

- (16) a. A: Sue wouldn't have kissed Peter last night.
 B: Well, who WOULD have (done)?
 b. If even Sue wouldn't have kissed Peter, I don't know who WOULD have (done)!

Lewis (2024:203) reports that long distance *wh*-subject extraction is impossible from within the BDE site (17), which, of the four analyses outlined in §2.1, only hers predicts. Interestingly, my consultants report that the acceptability of the examples in (17) improves significantly when *have* is the introducer verb (18), just like with the short distance *wh*-subject extraction cases:

- (17) I don't know who Tom thinks will leave, but I do know who Emma thinks will (*do).
 (18) I don't know who Tom thinks will have left, but I do know who Emma thinks will have (done).

The *Wh*-subject experiment aimed to determine whether my consultants' informal judgments regarding (16) to (18) aligned with those collected under experimental conditions from a more representative sample of the population. This experiment had 3 two-levelled factors, namely ELLIPSIS (*BDE*, *PAE*), LENGTH (*long*, *short*) AUX (*modal*, *perfect*), where LENGTH refers to long or short distance *wh*-subject extraction from within the ellipsis site and AUX refers to whether perfect *have* or a modal is employed as the introducer verb. ELLIPSIS was included to provide a benchmark, as nobody disputes that both long and short *wh*-subject extraction from PAE is acceptable, and that PAE is acceptable with either a modal or *have* as the introducer verb. The experiment tested for the possible outcomes that:

- (19) PO1: long-distance *wh*-subject extraction is prohibited from within the BD ellipsis site (following Lewis 2024)
 PO2: *wh*-subject extraction from the BD ellipsis site is preferred when *have* rather than a modal is the introducer verb (following my consultants' and Thoms' judgments)
 PO3: *wh*-subject extraction from the BD ellipsis site is preferred when *have* rather than a modal is the introducer verb, but only when extraction is either long- or short-distance.

Sample test items for each condition are presented in (20).

- (20) a. She won't say who Peter thinks should leave, but she will say who Emma thinks should (do). [long, modal]
 b. Who did Carol predict will have cried at the wedding? And who did she predict won't have (done)? [long, perfect]
 c. Were you told who might not go to New York? And, more importantly, were you told who might (do)? [short, modal]
 d. We don't know who Joe hopes will have baked a cake, but we do know who Jenna hopes will have (done). [short, perfect]

56 participants took part in this experiment, with 3 identified as outliers. Judgments from 53 participants were therefore included in the analysis, meaning 159 observations per condition. The mean ratings for the test items and standardized fillers are plotted in Figure 3.

Figure 3. Mean ratings for test and standardized filler items for the *Wh*-subject experiment

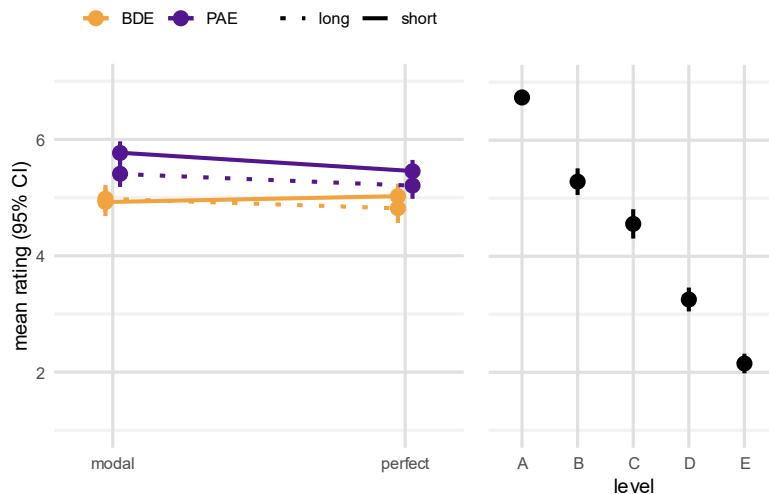


Figure 3 shows that, although there are slight preferences for short *wh*-extraction and using only a modal verb in the PAE condition, the four BDE conditions converge on the same rating, namely ~4.9 out of 7, which suggests that neither LENGTH nor AUX affects judgments on *wh*-subject extraction from a BD elliptical clause.

A summary of the relevant fixed effect estimates for the best fitting CLMM for the *Wh*-subject data is presented in Table 5. This summary shows that a significant main effect of ELLIPSIS is observed, which confirms that, generally speaking, both long- and short-distance *wh*-subject extraction is preferred with PAE to BDE. But because there are no significant two-way or three-way interaction effects, none of three possible outcomes listed above transpired (precisely which interaction effect tests for which possible outcome is shown in Table 5). This result is particularly troubling for Lewis' (2024) phase-based analysis of English predicate ellipsis (see §2.1), which predicts a significant contrast between short and long distance *wh*-subject movement, but only in BDE configurations.

Table 5. Summary of interaction model for the *Wh*-subject experiment

score ~ ellipsis * length * aux + (1 item) + (ellipsis subject)					
Contrasts	Estimate (β)	SE	z	p	Outcome tested for
<i>PAE</i> – <i>BDE</i>	0.91	0.18	5.11	<0.01	
<i>short</i> – <i>long</i>	0.27	0.15	1.75	0.08	
<i>modal</i> – <i>perfect</i>	0.28	0.15	1.84	0.07	
(<i>PAE_short</i> – <i>PAE_long</i>) – (<i>BDE_short</i> – <i>BDE_long</i>)	0.43	0.31	1.41	0.16	PO1
(<i>PAE_modal</i> – <i>PAE_perf</i>) – (<i>BDE_modal</i> – <i>BDE_perf</i>)	0.42	0.31	1.37	0.17	PO2
((<i>PAE_short_modal</i> – <i>PAE_long_modal</i>) – (<i>BDE_short_modal</i> – <i>BDE_long_modal</i>)) – ((<i>PAE_short_perf</i> – <i>PAE_long_perf</i>) – (<i>BDE_short_perf</i> – <i>BDE_long_perf</i>))	0.56	0.61	0.92	0.36	PO3

At this juncture, it is informative to compare results across the *Wh*-subject, Topicalization, and *Wh*-object experiments. To determine whether such a comparison is viable, one must first determine whether any significant differences obtain between ratings for the standardized fillers used in each experiment (recall that all three experiments used the same fillers). If no difference obtains, then a comparison is indeed viable. To undertake this cross-experiment comparison, one must switch to z-scored mean ratings, where the test item z-scores are calculated with reference only to the standardized fillers (which were identical in every experiment), as opposed to all experimental items, following Gerbrich et al. (2019). An intercepts-only LMM was fit to the standardized fillers from these three experiments, and indeed, no significant differences were observed, thus making a comparison viable.¹¹

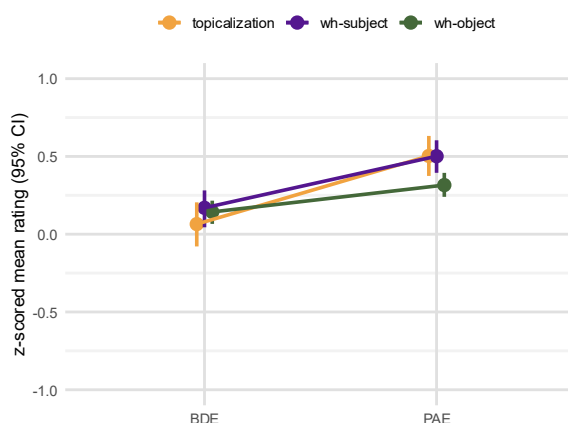
The comparison undertaken involved only a subset of the data, namely only the *short* condition test items from the *Wh*-subject experiment (where the *AUX* factor was averaged over), and the *inside* condition test items from the Topicalization and *Wh*-object experiment. More concretely, the comparison compared the configurations exemplified in (21).

- (21) a. We hope that Joe knows who will bake a cake, and that he also knows who won't (do). *wh-subject*
b. Jason's song I won't sing, but Ava's song I might (do). *topicalization*
c. I know which shoes I can reserve for Steve, and also which ones I can't (do). *wh-object*

As Figure 4 shows, short *wh*-subject extraction, *wh*-object extraction, and object topicalization all yield the same mean z-scored ratings in the *BDE* condition. My interpretation of this result is that each of these movement dependencies is compatible with *BDE*—these configurations are acceptable, in absolute terms. This interpretation is motivated by (i) the near unanimous agreement in the literature that short *wh*-subject extraction in a *BD* elliptical clause is perfectly acceptable (only Griffiths & Den Dikken 2022 suggest otherwise), and (ii) the degree of degradation observed for the *BDE* configurations in Figure 4 (when compared to their *PAE* counterparts) is mild when compared to the degree of degradation that will be observed for *BDE* in the two experiments discussed later in §3, namely the Passivization and Copula experiments. In those experiments, the average score for *BDE* falls within the 3–4 range, as opposed to falling within the 4–5 range, as observed here.

¹¹ The z-scored mean ratings for the standardized fillers were coded for their originating EXPERIMENT (*top*, *wh-subj*, *wh-obj*) and compared level-by-level (i.e., using the formula “zscore ~ experiment * level + (1 | item)”, with repeated-contrast coding for the fixed effect *level*) and no interaction effects were observed. Note that using z-scored mean ratings is justified in this instance because only minimum scale distortion was observed in the experiments being compared, as the distributions of standardized items in Figures 1 to 3 show.

Figure 4. Z-scored mean ratings for the [+inside] subset of the test items for the *wh*-subject, Topicalization, and *Wh*-object experiments



The observation that *wh*-object extraction and object topicalization return almost identical acceptability ratings in the *BDE* condition seriously undermines Lewis' (2024) and Thoms & Sailor's (2018) analyses, which, for different reasons (see §2.1), both predict a measurable difference in acceptability between these conditions.

2.5. Relativization and comparatives

Seeing as the four exemplar analyses from §2.1 make different claims and predictions about the compatibility of *BDE* with overt relativization and covert operator movement, it appears crucial to obtain reliable judgments on *BDE* in relative and comparative clauses, in which these movement operations apply.

The empirical picture regarding *A'*-extraction from within the *BDE* site in relative clauses is rather complicated. Thompson (2014) claims that restrictive relatives differ in their compatibility with *BDE* according to whether the elliptical relative clause is contained in its own antecedent (so-called *antecedent contained deletion*, *ACD*) or not:

(22) *Standard restrictive relative clause*

- * I sold the furniture I knew my cat might scratch, and I kept the pieces that he already had done. (Thompson 2014:46)

(23) *Antecedent contained restrictive relative clause*

- He has read every book he must do. (Thompson 2014:45)
- They raise no interpretative difficulties of the kind that credo may do. (BNC David Johnston, *The Roman Law of Trusts*; Thompson 2014:110)
- He could not feel the same bitterness that he had done when he first started to write. (BNC Helen Forrester, *The Latchkey Kid*; Thompson 2014:110)
- A man who steals does not incur the same measure of public reprobation which he would have (done) in the past. (cited from a speech on policing; see Baker 1984)

Thoms & Sailor (2018) propose that *A'*-extraction from within the *BD* ellipsis site in relative clauses is permitted only if the relative clause can be derived via matching (Chomsky 1977, see the abstract schema in (24a)). This proposal overlaps to a significant degree with Thompson's generalization, as *ACD* configurations are generated by matching (Fox 2002). However, there is some friction here: Abels (2012) and Thoms & Sailor (2018) report the examples in (25) as deviant / unacceptable with *BDE* and suggest that they are problematic because amount-relatives and headless free relatives can only be generated by raising (see the schema in (24b)), yet both examples involve *ACD* and therefore should only be derivable by matching and should be acceptable.

- (24) a. [XP-HEAD_i [REL-CLAUSE Op_i1 [... t₁ ...]]] *basic matching configuration*
b. [REL-CLAUSE XP-HEAD₁ [... t₁ ...]] *basic raising configuration*

- (25) a. I put in my pocket all the money I could (??do).
b. He buys what he can (*do). (Abels 2012:32)

This picture is further blurred by ACD relativization data from Chalcraft 2006, Haddican 2007, and Aelbrecht 2010, which are reported as unacceptable (26).

- (26) a. * Bart can eat anything that Homer can do. (Haddican 2007:540)
 b. * John hit every target Mary did do. (Chalcraft 2006:10)
 c. ?* Ed must read every article he can do. (Aelbrecht 2010:360)

Schwarzer (2023) reports that ACD relativization is only acceptable with BDE when the subject of the relative clause is a bound pronoun, see (27). This implies that, for her consultants, (23a,c,d), (26c), and (25) are acceptable, whereas (23b), (26a), and (26b) are unacceptable.

- (27) Jane_i will read every book that {she_i / *Sue} should do.

There is agreement in the literature that BDE is acceptable in comparative clauses such as (28). For Thompson and Thoms & Sailor, this outcome is unsurprising, as these elliptical clauses are contained in their own antecedents and are widely viewed as derived by matching (Kennedy & Merchant 2000, Lechner 2004).

- (28) a. John stole more money than he should have done. (Baker 1984:156)
 b. At first he felt more relaxed than he had done in a long time. (Thompson 2014:45)
 c. Had Blader taken more champagne on board than he should have done? (Algeo 2006:290)

I conducted two experiments on relative and comparative BDE configurations. The first, the Standard-rel experiment, tested whether Thompson's report about A'-movement from the BDE site being prohibited in standard relative clauses (recall (22)) is representative. Because my consultants reported a small improvement in acceptability with *done* instead of *do* for such configurations (compare (29a) and (29b)), this experiment included the factor AUX (*modal*, *perfect*) in addition to the two factors tested in the previous two experiments, namely ELLIPSIS (*BDE*, *PAE*) and POSITION (*inside*, *outside*).

- (29) a. Sue was offered one job which she might not start right away, and another which she might *do*.
 b. Sue was offered one job which she might not have started right away, and another which she might have *done*.

The possible outcomes tested for in the Standard-rel experiment were:

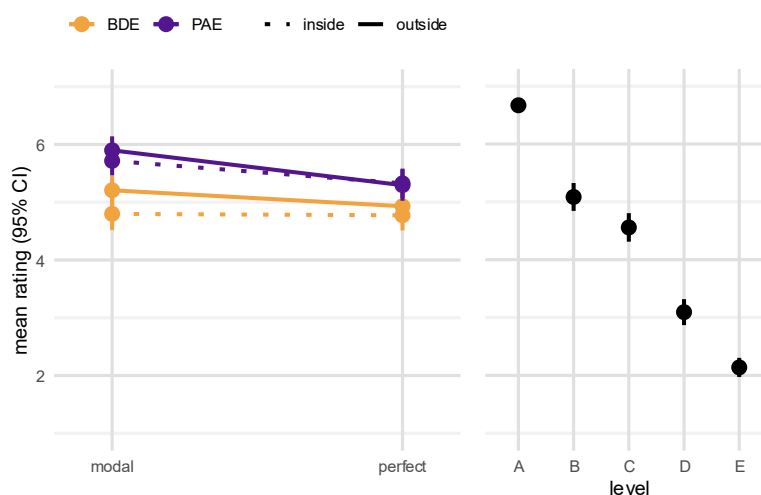
- (30) PO1: A'-movement from the BD ellipsis site in standard relative clauses is prohibited (following Thompson 2014).
 PO2: For both BDE and PAE, having *have* rather than a modal as the introducer verb yields an amelioration effect on A'-movement from within the ellipsis site.
 PO3: Having *have* rather than a modal as the introducer verb yields an amelioration effect for BDE but not PAE (following my consultants).
 PO4: Having *have* rather than a modal as the introducer verb yields an amelioration effect for BDE but not PAE, but only when direct object relativization occurs.

The relative clauses in the *inside* condition test items displayed direct object relativization (see (29) for example items), whereas those in the *outside* condition test items displayed subject relativization (see (31)). All test items displayed an overt element in the complementizer domain of the relative clause. A third of the test items displayed a bare *wh*-word (e.g., *who*), a third displayed a complex *wh*-phrase (e.g., *which student*), and a third displayed the complementizer *that*.

- (31) a. Mike changed one light bulb which might not break and another which might (do).
 b. Mike changed one light bulb which might not have broken and another which might have (done).

56 participants took part in this experiment, with 3 identified as outliers. Judgments from 53 participants were therefore included in the analysis, meaning 159 observations per condition. The mean ratings for the test items and standard fillers are plotted in Figure 5.

Figure 5. Mean ratings for test and standardized filler items for the Standard-rel experiment



A summary of the fixed effect estimates for the best fitting CLMM for the Standard-rel data is presented in Table 6. This model returns significant main effects for all three factors and a significant interaction effect that appears to support Possible Outcome PO3 from (30) but actually does not, as the superadditive effect goes in the wrong direction to support PO3. Considering that none of the four possible outcomes from (30) transpired, one can conclude that A'-movement from the BDE site in standard relative clauses is permitted (*contra* Thompson 2014 and the predictions of Aelbrecht 2010 and Baltin 2012), and that there is no BDE-specific preference for employing *have* rather than a modal as the introducer verb (*contra* my consultants). Anecdotal support for this conclusion comes from existence of the naturally-attested examples such as (32), which involves relativization from within the BDE site to form a standard restrictive relative clause (albeit with ATB-movement):

- (32) Then Toby realized that Shorty was sitting beside him. And that Shorty must have been hovering all the time in the toilet at the back of the café, which was *something Toby hadn't thought of and should have done*, but clearly Shorty had.
(from John le Carré's *A Delicate Truth*, cited in Thompson 2014:240)

Table 6. Summary of interaction model for the Standard-rel experiment

score ~ ellipsis * length * aux + (1 item) + (ellipsis subject)					
Contrasts	Estimate (β)	SE	z	p	Outcome tested for
<i>PAE – BDE</i>	1.23	0.20	6.12	<0.01	
<i>outside – inside</i>	0.30	0.15	2.04	0.04	
<i>modal – perfect</i>	0.65	0.15	4.34	<0.01	
<i>(PAE_out – PAE_in) – (BDE_out – BDE_in)</i>	–0.09	0.30	–0.32	0.75	PO1
<i>(in_modal – in_perf) – (out_modal – out_perf)</i>	0.41	0.30	1.38	0.17	PO2
<i>(PAE_modal – PAE_perf) – (BDE_modal – BDE_perf)</i>	0.79	0.30	2.66	0.01	PO3
<i>((PAE_modal_out – PAE_perf_out) – (BDE_modal_out – BDE_perf_out)) – ((PAE_modal_in – PAE_perf_in) – (BDE_modal_in – BDE_perf_in))</i>	0.06	0.59	0.10	0.92	PO4

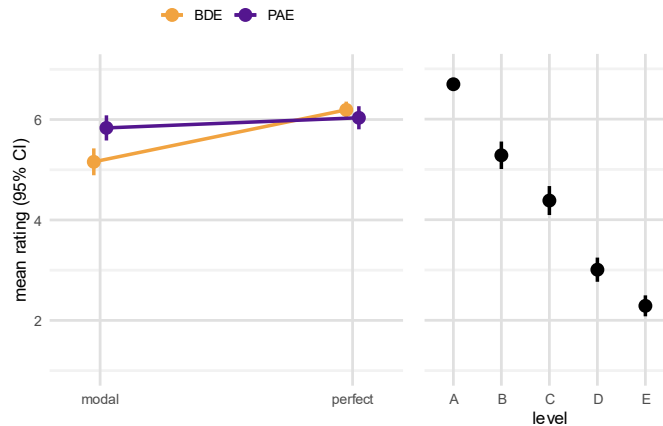
The second experiment conducted on BDE in relativized subclauses, called **ACD**, focused on elliptical clauses that are contained in their own antecedents, i.e. ACD configurations. In preparation for this experiment, I asked my consultants about Thompson's examples from (23) and variations thereon. They judged such examples as acceptable, but again reported a preference for using perfect *have* rather than a modal as the introducer verb. The **ACD** experiment aimed to discover if their judgments are replicated in a larger sample of British English speakers. The experiment included two factors, **ELLIPSIS** (*BDE*, *PAE*) and **AUX** (*modal*, *perfect*), and tested the claim that BDE in ACD configurations is preferred with *have* as the introducer verb (relative to PAE). Note that the factor **POSITION**, which was included in the Standard-rel experiment, cannot be included here, as ACD relatives necessarily involve A'-extraction from the ellipsis site.

Due to a mistake in the experimental design, the **ACD** experiment was actually conducted as two separate subexperiments, one that tested elliptical ACD comparative clauses (henceforth *ACD_{comp}*), and another that tested a balanced mixture of configurations whose ACD-subclause is a comparative, *that*-relative, or headless relative clause (henceforth *ACD_{mix}*). In *ACD_{comp}*, each list contained 7 items in which the elliptical comparative clause's subject coreferred with the matrix clause's subject (henceforth *coref*) and 5 items in which this subject did not corefer with the matrix subject (henceforth *disjoint*). In *ACD_{mix}*, all elliptical subclauses had *coref* subjects. Example test items for both subexperiments are presented in (33).

- (33) a. i. Rosamund will read more books this week than Ben will (do). [modal]
 ii. Laura won't cook more pasta than she should (do).
 iii. Anna will sing every song that she can (do).
 iv. Helen didn't bring what she said she would (do).
 b. i. Ant won't have earned as much money as Dec will have (done). [perfect]
 ii. Liv laughed louder than she should have (done).
 iii. Bob fixed all the problems that he could have (done).
 iv. Fin hasn't sold what he could have (done).

40 participants took part in the *ACD_{comp}* subexperiment, with 2 identified as outliers. Judgments from 38 participants were therefore included in the analysis, meaning 114 observations per condition. The mean ratings for the test items and standardized fillers are plotted in Figure 6.

Figure 6. Mean ratings for test and standardized filler items for the *ACD_{comp}* subexperiment



A summary of the fixed effect estimates for the best fitting CLMM for the *ACD_{comp}* data is presented in Table 7. This summary shows significant main effects for *ELLIPSIS* and *AUX* and a significant interaction effect. The presence of a significant interaction confirms that, on this occasion, my consultants' report that *done* is preferred over *do* in BDE-ACD configurations is representative. However, based on the high mean scores for both BDE conditions in Figure 6 (between standardized filler levels A and B), I interpret this result as demonstrating that A'-extraction from within BDE ellipsis site in an ACD configuration is grammatically licit but that BDE is subject to a non-syntactic (probably prosodic) constraint that is satisfied in the *perfect* condition but not in the *modal* condition. Additional support for this interpretation will be presented in §3.2, when the results of the *COPULA* experiment are reported. There, we will see that *done* (as opposed to *do*) ameliorates acceptability, but not by the amount expected if the source of the *do-done* modulation is syntactic in nature.

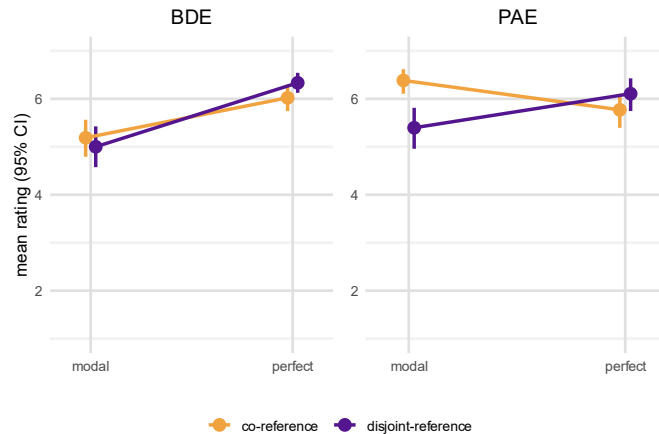
Table 7. Summary of interaction model for the *ACD_{comp}* subexperiment

score ~ ellipsis * aux + (1 item) + (ellipsis subject)				
Contrasts	Estimate (β)	SE	z	p
PAE – BDE	0.81	0.35	2.28	0.02
modal – perfect	–1.28	0.35	–3.71	< 0.01
(PAE_modal – PAE_perf) – (BDE_modal – BDE_perf)	–1.64	0.68	–2.40	0.02

One might wonder whether having a *disjoint* (e.g., (33i)) or *coref* (e.g., (33ii)) subject in the elliptical comparative clause affects acceptability. To investigate this, I conducted a post-hoc analysis that involved extracting the *disjoint* test items (3 repetitions of 5 lexical sets) and comparing them to an equal

number of randomly selected *coref* test items. Figure 7 shows that, although modulating the status of the elliptical clause's subject has a noteworthy effect on the acceptability of PAE (especially in the *modal* condition), modulating the subject's status has a negligible effect on the acceptability of BDE. Statistical testing confirms this impression. The best fitting CLMM for only the BDE data (namely, subject-type * aux + (1 | item) + (aux | subject)) returns no significant main effect for SUBJECT-TYPE ($\beta = -0.27$, $z = -0.59$, $p = 0.55$), but a significant main effect for AUX ($\beta = -2.21$, $z = -4.28$, $p < 0.01$). No significant interaction effect is observed ($\beta = -1.13$, $z = -1.16$, $p = 0.25$).

Figure 7. Mean ratings for a subset of test items from the ACD_{comp} subexperiment

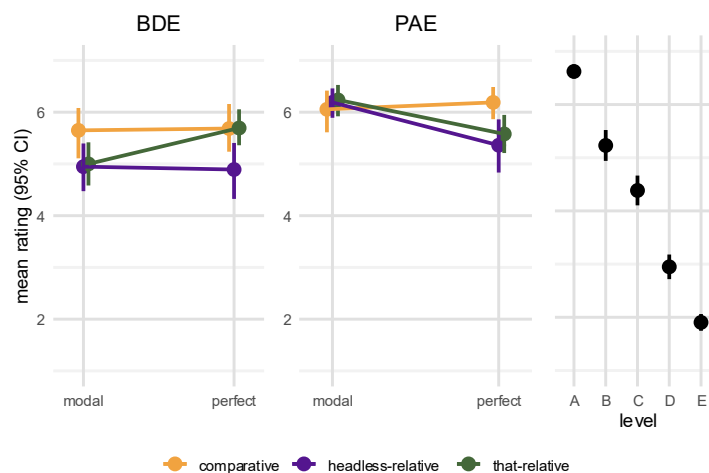


Recall that Schwarzer (2023) reports that ACD *that*-relatives such as (27) (repeated in (34)) are compatible with BDE only when the subject of the elliptical clause is a bound pronoun, i.e. *coref*. Because Schwarzer does not mention whether the variation in acceptability reported in (34) is also observed in comparatives, it is impossible to know whether or not the results in Figure 7 represent a refutation of Schwarzer's claims. At the very least, these results show that the variation reported in (34) is not observed in *all* ACD configurations displaying BDE.

(34) Jane_i will read every book that {she_i / *Sue} should do.

Because it contains a balanced mix of types of ACD subclauses, one can view the ACD_{mix} subexperiment as containing an additional factor, namely SUBCLAUSE-TYPE (levels = *comparative*, *headless relative*, *that-relative*). 39 participants took part in the ACD_{mix} subexperiment, with 2 identified as outliers. Judgments from 37 participants were therefore included in the analysis, meaning 37 observations per condition. The mean ratings for the test items and standardized fillers are plotted in Figure 8.

Figure 8. Mean ratings for test and standardized filler items for the ACD_{mix} subexperiment



The results in Figure 8 show that, in the *perfect* condition, roughly equivalent ratings are returned for each type of ACD configuration regardless of whether BDE and PAE applies. For instance, the

headless relative configurations (e.g., (33biv)) receive an average rating of 4.89 in the *BDE* condition and 5.36 in the *PAE* condition, which is a difference of only ~0.5 points.

Due to the small number of observations obtained for each condition in the ACD_{mix} subexperiment, one must treat any statistical modelling of this subexperiment's results as preliminary, that is to say, as indicative only of possible trends. With this in mind, an intercepts-only CLMM was fit to a subset of the ACD_{mix} data (namely, only items in the *perfect* condition). In this model, ELLIPSIS was treatment-coded, and SUBCLAUSE-TYPE was difference-coded. According to this model, in the *BDE* condition, insignificant differences in acceptability obtain between ACD configurations with comparative clauses and with headless relative clauses ($\beta = 1.48$, $z = 1.68$, $p = 0.21$), and between ACD configurations with headless relative clauses and with *that*-relative clauses ($\beta = 1.03$, $z = 1.14$, $p = 0.49$). Furthermore, the presence of BDE or PAE yields no significant differences in acceptability for any subclause-type (for comparatives, $\beta = 0.75$, $z = 0.84$, $p = 0.40$; for headless relatives, $\beta = 0.75$, $z = 0.86$, $p = 0.38$; for *that*-relatives, $\beta = 0.09$, $z = 0.10$, $p = 0.92$).

In the literature, PAE is viewed as acceptable with each of the three ACD configurations tested in this subexperiment. Considering that the results from ACD_{mix} suggest that BDE and PAE return the same ratings for each ACD configuration, one must tentatively conclude that BDE is compatible with all ACD configurations. This suggests that Abels' (2012) and Thoms & Sailor's (2018) claim that BDE is incompatible with headless relative clauses is incorrect.

2.6. British English *do* in pseudogapping configurations

As mentioned in §1, Pullum & Wilson (1977) report the cooccurrence of British English *do* and pseudogapping as acceptable (Thoms (2011b) agrees), whereas Baltin (2012) reports it as unacceptable:

- (35) a. Chomsky revolutionized English phonology, as earlier he had done English syntax.
(Pullum & Wilson 1977:762)
b. * Although he wouldn't visit Martha, he would do Sally. (Baltin 2012:391)

Each of my five consultants judged the examples in (35) as acceptable. If British English *do* is indeed permitted in pseudogapping configurations, then the phrase targeted by ellipsis in pseudogapping is either the same size as the phrase targeted in standard BDE configurations or smaller, depending on whether one assumes that the complement of *do* is elided and that the remnant of pseudogapping moves to a right-adjoined position (see (36a) and §5.1), or one assumes that the remnant of pseudogapping must leftward move to a position above the ellipsis site to escape deletion (see (36b); Gengel 2007 and references therein). This implication conflicts with the prevailing view, originating with Merchant 2008a, that ellipsis targets a larger phrase in pseudogapping than in both standard PAE and BDE. It also undermines Aelbrecht's (2010) and Baltin's (2012) derivational-timing analyses of BDE (see §2.1), which both subscribe to the prevailing view.

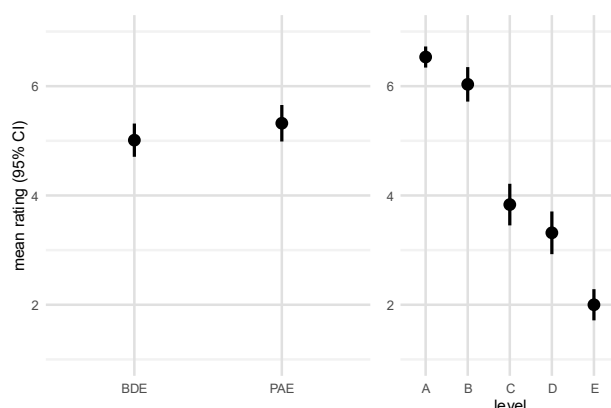
- (36) a. ... as earlier he had [done [_{XP} revolutionized *t*₁] [English syntax]₁]
b. ... as earlier he had [done [_{VP} [English syntax]₁ Y⁰ [_{XP} revolutionized *t*₁]]]

The Pseudogap experiment tested for the possibility that pseudogapping is just as acceptable with BDE as with PAE. This experiment had 1 two-levelled condition, namely ELLIPSIS (*BDE*, *PAE*). Two sample test items are provided in (37). Note that all BDE test items used *do*; none used *done*. Note also that none employed an *as*-parenthetical clause, as in (35a).

- (37) a. The fire spread quickly! Apparently, the firewoman was able to help John sooner than she could (do) Paula.
b. The boss is being unfair to his administrative staff. In my view, he should treat his secretary just like he would (do) any other employee.

30 participants took part in this experiment, with 2 identified as outliers. Judgments from 28 participants were therefore included in the analysis, meaning 84 observations per condition. The mean ratings for the test items and standard fillers are plotted in Figure 9.

Figure 9. Mean ratings for test and standardized filler items for the Pseudogap experiment



The best fitting CLMM that modelled ELLIPSIS as a fixed effect included random intercepts for items and random slopes for subjects. No significant simple effect for ELLIPSIS was observed ($\beta = -0.79$, $z = -0.79$, $p = 0.43$). This result aligns with the above-stated prediction that pseudogapping is compatible with BDE (*contra* Baltin 2012) and strongly suggests that my five consultants' judgments are representative.

2.7. Summary of the experiments on the movement restrictions on BDE

Table 8 summarizes the relevant findings from the experimental results reported in this section. In this table, '(in)compatibility' refers to whether the interaction of two phenomena causes a grammatical (i.e., syntactic) violation or not, and the amelioration effect of perfect *have* should be understood as relative to having a modal as the introducer verb.

Table 8. Summary of the relevant findings of six acceptability judgment experiments on the movement restrictions on BDE

Experiment	Main finding	Ameliorating effect of perfect <i>have</i> on BDE (but not PAE)?
<i>Wh</i> -subject	BDE is compatible with both long- and short-distance <i>wh</i> -subject extraction from the elliptical clause	no
Topicalization	BD is compatible with topicalization from within the ellipsis site	not tested
<i>Wh</i> -object	BD is compatible with <i>wh</i> -object extraction from within the ellipsis site	not tested
Standard-rel	BD is compatible with object relativization from within the ellipsis site in a standard relative clause	no
ACD	BD is compatible with object relativization from within the ellipsis site in an ACD subclause	yes
Pseudogap	BDE is compatible with pseudogapping	not tested

One might extrapolate from Table 8 that there are no unique restrictions on BDE whatsoever. This is untrue, however. In the next section, I propose that the BDE fits the *Raising Verb Averse* generalization, which states that the BDE site cannot be headed by a raising verb, that is to say, a verb capable of raising to T. In §4, I return to the four exemplar analyses of BDE originally outlined in §2.1, in order to critically assess their ability to capture the findings summarized in Table 8 and the new generalization from §3. Because no extant analysis can capture the updated BDE dataset, I offer in §5 a novel analysis that does.

3. A novel generalization about BDE (and two more experiments)

The experiments reported in §2 mostly returned null results, which I interpreted as demonstrating that there are no movement restrictions on BDE, contrary to the consensus position in the literature. From this, the question arises whether BDE differs formally from PAE whatsoever (aside from the obvious addition of *do*). I claim that BDE is indeed more restricted than PAE. Specifically, I claim that the following, previously unnoticed generalization holds:

(38) *The Raising Verb Averse generalization*

British English *do* cannot immediately precede an ellipsis site if that ellipsis site is headed by a raising verb.

In (38), the term ‘raising verb’ refers to any verbs that, when structurally closest to T in a finite clause, would undergo head-movement to T. Preliminary support for (38) comes from the following data, which report the informally collected judgments of my 5 native-speaking consultants.

BD is incompatible with raising possessive-*have* (39), and with ellipsis sites headed by copula *be* (40). My consultants agree with Thoms & Sailor (2018) and Lewis (2024) that BD is compatible with ellipsis sites headed by passive-*get*, which is a non-raising verb, but not with ellipsis headed by passive-*be*, which is a raising verb.

(39) John hasn’t any money left, and Mary hasn’t (*done), either.

(40) Clara might be a teacher, and Sam might (*do), too.

- (41) a. The cookies definitely won’t *get* eaten, but the cakes might (do).
b. The cookies definitely won’t *be* eaten, but the cakes might (*do). (Thoms & Sailor 2018:5)

Sailor (2014:30–33) shows that the obligatorily perfect idiomatic phrase *have been PP_{to}* (meaning ‘visited’) cannot be recovered for the PAE site in (42): these ellipsis sites can only be recovered as *will go there*. Because the elliptical VP and its antecedent therefore mismatch (*be* versus *go*), the elliptical utterance is degraded, in terms of acceptability. The same degradation in acceptability is reported by my consultants for the BDE counterpart in (42b), which demonstrates that the BD ellipsis site cannot be headed by perfect *have*, which is a raising verb.¹²

(42) John will have been to that new leather bar, and ...

- a. #... Bill will [have been there], too.
b. #... Bill will do [have been there], too.

(43) demonstrates that *dying to* is an obligatorily progressive idiom chunk, as its idiomatic interpretation (namely, ‘want very much to’) is lost when *die* does not arise in its present participle form. The idiomatic interpretation of this phrase is retained under PAE when progressive *be* is seemingly absent (44a), which shows that *be* is actually contained in the ellipsis site (44b) (Sailor 2014).

(43) *Obligatorily progressive* (Sailor 2014:30)

- a. John {was / has been / will be} *dying to* go to that new leather bar.
b. #John died to go to that new leather bar. (idiomatic reading lost)
c. #John has died to go to that new leather bar. (idiomatic reading lost)

- (44) a. John will be *dying to* go, and Mary will, too. (idiomatic reading retained)
b. ... Mary will *be dying to* go, too.

The idiomatic interpretation is unavailable under BDE however: (45a) can only mean something odd such as ‘Mary will die, too’. One concludes from this that the BDE site cannot be headed by progressive *be*, which is another raising verb (45b).

- (45) a. John will be *dying to* go, and Mary will do, too. (idiomatic reading lost)
b. *... Mary will do *be dying to* go, too.

Although these informally collected datapoints are encouraging for the Raising Verb Averse generalization, each requires experimental confirmation before the generalization can be said to reliably hold. Unfortunately, due to limited resources, I was only able to experimentally confirm two claims, namely that the BDE site cannot be headed by copula or passive *be*. These experiments are reported in the following two subsections.

¹² Thoms (2019:1028) reports that some speakers permit PAE of verbal phrases headed by perfect *have*, and that this variation is idiolectal in nature. Whether those speakers who permit PAE in such configurations also permit BDE remains to be determined. Each of my 5 consultants disliked both (42a) and (42b).

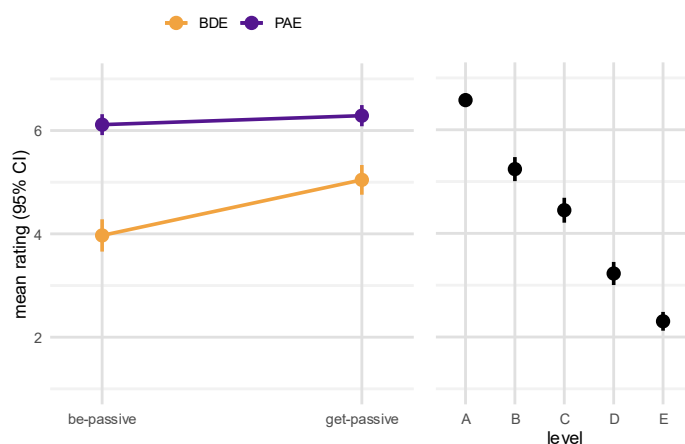
3.1. Be- versus get-passives

The Passivization experiment tested the claim from Thoms & Sailor (2018), Lewis (2024), and my consultants that BDE is less acceptable with *be*-passives than with *get*-passives. To demonstrate that this pattern is not a general property of English predicate ellipsis but is indeed unique to BDE, passivization with PAE was included in the experimental design. Based on the undisputed assumption that PAE is acceptable with both *be*- and *get*-passives, the abovementioned claim is supported only if an interaction effect is observed between the two factors ELLIPSIS (*BDE*, *PAE*) and PASSIVE (*be*, *get*). Sample test items for each condition are presented in (46).

- (46) a. The cake won't be eaten, but the cookies might (do). *be*-passive
b. The cake won't get eaten, but the cookies might (do). *get*-passive

47 participants took part in this experiment, with 3 identified as outliers. Judgments from 44 participants were included in the analysis, meaning 132 observations per condition. The mean ratings for the test items and standardized fillers are plotted in Figure 10.

Figure 10. Mean ratings for test and standardized filler items for Passivization experiment



As expected, both types of passive received near-ceiling scores with PAE. Switching to BDE lowers scores for both types of passive but lowers scores to a greater extent with *be*-passives, as predicted. The absolute mean rating attributed to the *BDE_be* condition, which is notated using the unacceptability asterisk in the literature, is ~4 out of 7.

A summary of the fixed effect estimates for the best fitting CLMM for the Passivization data is presented in Table 9. This summary shows that a significant interaction effect is observed, thus supporting the claim that *get*-passives are more acceptable than *be*-passives, but only in BDE configurations, which provides experimental support for the Raising Verb Averse generalisation in (38). Significant main effects of ELLIPSIS and PASSIVE are also observed, which confirms that, generally speaking, passivization is preferred with PAE over BDE, and *get*-passives are preferred to *be*-passives, when one averages over PAE and BDE.

Table 9. Summary of interaction model for the Passivization experiment

score ~ ellipsis * passive + (1 item) + (ellipsis subject)				
Contrasts	Estimate (β)	SE	z	p
<i>PAE – BDE</i>	3.71	0.45	8.24	<0.01
<i>get – be</i>	1.13	0.28	4.05	<0.01
<i>(PAE_get – PAE_be) – (BDE_get – BDE_be)</i>	–1.12	0.55	–2.02	0.04

3.2. BDE of copular clauses

Recall from §3.1 that my consultants judged BDE as unacceptable with copular phrase ellipsis sites, which are headed by the raising verb *be* (47). Not mentioned in §3.1 was that they also reported a slight improvement with such configurations when the introducer verb was perfect *have* rather than a modal verb (48).

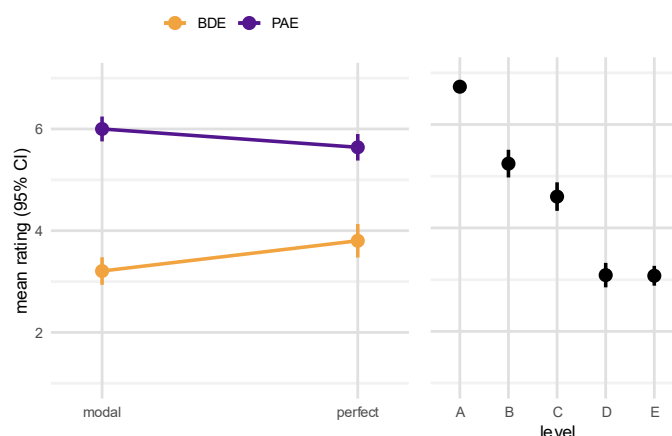
- (47) a. Clara might be a teacher, and Sam might (*do) [be a teacher], too.
 b. Nathan won't be interested, but Rachel will (*do) [be interested].
- (48) a. Harry should be more considerate, and Penny should (*do), too.
 b. Harry should have been more considerate, and Penny should have (?done), too.

The Copula experiment aimed to determine whether my consultants' informal judgments aligned with those collected under experimental conditions from a more representative sample of the population. This experiment had 2 two-levelled factors, namely ELLIPSIS (*BDE*, *PAE*) and AUX (*modal*, *perfect*),¹³ and tested the claims that (i) copular phrase ellipsis sites are less acceptable with BDE than PAE, and (ii) having perfect *have* rather than a modal as the introducer verb ameliorates degradation in acceptability with BDE but not with PAE. Sample test items for each condition have already been presented in (48). Note that nobody disputes the acceptability of PAE with copula ellipsis sites (49) (cf. Lipták & Merchant 2019) (provided that copula *be* arises in the same inflectional form in the antecedent and elliptical clauses; see Warner 1985 for details).

- (49) a. Abby should be grateful, and Tom should [be grateful], too.
 b. Olivia could be a singer one day, whereas Taylor couldn't [be a singer].

40 participants took part in this experiment, with 3 identified as outliers. Judgments from 37 participants were included in the analysis, meaning 111 observations per condition. The mean ratings for the test items and standardized fillers are plotted in Figure 11.

Figure 11. Mean ratings for test and standardized filler items for the Copula experiment



Despite not reaching the ceiling value, the mean ratings for test items in the PAE condition are high enough to declare with confidence that PAE with a copular phrase ellipsis site is acceptable, as expected. The results show that BDE with a copular phrase ellipsis site is strongly degraded (down to the standardized filler “D” and “E” levels), with the presence of perfect *have* immediately preceding British English *do* (as opposed to a modal verb) providing an amelioration effect, again as predicted.

A summary of the fixed effect estimates for the best fitting CLMM for the Copula data is presented in Table 10. This summary shows that a significant main effect of ELLIPSIS is observed, which confirms that, generally speaking, copular phrase ellipsis sites are preferred with PAE over BDE. A significant interaction effect is also observed, thus supporting the claim that an amelioration effect is provided in the BDE condition by using *have* as the introducer verb, as opposed to a modal.

¹³ Note that BD configurations in which copula *be* has raised to T were not included in the Copula experiment because there is no need to experimentally test them—they are undisputedly bad, yielding floor level judgments (unlike their acceptable PAE counterparts):

(i) John is a doctor and Pete is (*do), too.

The same pattern is observed with raising *have* but not with finite perfect *have* (compare (ii) and (iii)) in British English, as already mentioned in §3.1. The analysis presented in §5 will capture these facts.

(ii) John hasn't any money left, and Mary hasn't (*done), either.

(iii) Although Tara has finished the race, Ashley hasn't (done).

Table 10. Summary of interaction model for the Copula experiment

score ~ ellipsis * aux + (1 item) + (ellipsis subject)				
Contrasts	Estimate (β)	SE	z	p
<i>PAE – BDE</i>	4.44	0.49	8.98	<0.01
<i>modal – perf</i>	–0.04	0.25	–0.16	0.87
<i>(PAE_modal – PAE_perf) – (BDE_modal – BDE_perf)</i>	–1.85	0.52	–3.60	<0.01

As Figure 11 shows, modulating between *modal* and *perf* in the BDE condition does not return acceptability to the levels enjoyed by the PAE test items (i.e., between standardized filler levels A and B). I interpret this as demonstrating that the ameliorating effect of *have* has an independent source from the degrading effect of having a copula-containing ellipsis site in BDE. Specifically, I view these results as providing further evidence, beyond the evidence already returned by the ACD_{comp} experiment (see §2.5), that BDE's acceptability is contingent on independent prosodic and syntactic constraints being satisfied. Test items in the *BDE_modal* condition violate both the prosodic and the syntactic constraints, whereas test items in *BDE_perfect* condition satisfy the prosodic constraints but violate the syntactic ones. Acceptability rises only marginally when switching from the *BDE_modal* to the *BDE_perfect* condition because syntax has a much larger effect on acceptability than prosody. The precise nature of the syntactic and prosodic constraints involved is discussed in §5 and §6 respectively.

In summary, the results of the Copula experiment provide further experimental support for the generalization in (38), namely that the BDE site cannot be headed by a raising verb.

4. Assessing the viability of the previous analyses outlined in §2.1

Table 1 from §2.1 presented a summary of the stated and/or predicted movement restrictions on BDE according to four reasonably recent analyses. This table can now be updated to include the findings from the judgment experiments reported in the previous sections:

Table 11. Stated and/or predicted movement restrictions on BDE, according to four recent analyses (updated from Table 1 to include the results of the 8 experiments reported in §2–3)

Type of movement		stated / predicted compatibility with BDE				results from §2–3
		Aelbrecht 2010, Baltin 2012	Thompson 2014	Thoms & Sailor 2018	Lewis 2024	
overt A'-mvmt	<i>wh</i> -object	x	x	x	x	✓
	long distance <i>wh</i> -subject	✓	✓	✓	x	✓
	object topicalization	x	x	✓	✓	✓
	raising-style relativization	(x)	(x)	✓	(unknown)	✓
	pseudogapping	x	x	(unknown)	(✓)	✓
covert A'-mvmt	operator movement	x	✓	✓	(unknown)	✓
	quantifier raising	x	✓	✓	✓	not tested
overt A-mvmt	subject raising	✓	✓	✓	✓	not tested
	unaccusatives	✓	(x)	✓	✓	not tested
	<i>be</i> -passivization	x	x	[x]	x	x
	<i>get</i> -passivization	(x)	x	✓	✓	✓

Recall from §2.1 that each previous analysis was built around the assumption and/or claim that BDE displays at least some movement restrictions. The experimental results from §2 suggest that these assumptions and claims are incorrect, and that BDE is actually fully compatible with all forms of phrasal movement, just as PAE is. More precisely, the experimental results indicate that the BDE site is, syntactically speaking, fully permeable to extraction from within it, which severely undermines the analyses developed in Aelbrecht 2010, Baltin 2012, and Lewis 2024, according to which the BDE site is often spelled-out before certain types of moving phrases can escape it. The observations that (i) no differences in acceptability obtain between *wh*-subject, *wh*-object, and object topicalization BDE configurations and (ii) no meaningful differences in acceptability obtain for BDE across different subclauses (i.e., various types of relative or comparative clauses) or across movement types (e.g., overt versus covert movement) severely undermine Thompson's (2014) and Thoms & Sailor's (2018) analyses, which predict these differences to obtain. Furthermore, because no previous study has observed that BDE appears to fit the Raising Verb Averse generalization in (38), no previous analysis can account for it.

Although none of these analyses can capture the updated BDE dataset, the question remains whether the concepts underlying them should be used as building blocks for future analyses. In the remainder of this section, I present reasons for why certain components of these analyses should not feature in future attempts to capture the distribution of BDE.

Recall from §2.1 that Aelbrecht's and Baltin's analyses rested on the claim that PAE and BDE are different forms of predicate ellipsis, both in terms of the phrase targeted for deletion (vP in PAE and VP in BDE), and the syntactic licenser for ellipsis (Voice for PAE and v for BDE). However, the current evidence—some of which was admittedly collected informally and therefore awaits experimental confirmation—strongly suggests that the contrary holds. First, both BDE and PAE are licensed only when T contains a pronounced, non-contracted verb (King 1970, Baltin 2012:392, Griffiths & Den Dikken 2022:120).¹⁴

- (50) a. * Pete'll arrive late and John'll (do), too.
b. Pete'll have scored higher than 70 in the exam, I reckon. * John'll have (done), too.

Second, when the subject of the elliptical clause is 'unusual' (e.g., a subject of a specificational copular clause, or a null element in locative inversion), neither BDE nor PAE is licensed (Griffiths & Den Dikken 2022:121):

- (51) a. * For this theory, BOB's biggest worry will {seem / be likely} to be the agreement facts; for that theory, KATE's biggest worry will (do).
b. * The TRIGGER of the political protest might {seem / be likely} to be food shortages; ironically, the OUTCOME of the protest might (do), too.
c. * On THIS building would fly the Confederate flag; on THAT building would (do), too.

Third, neither PAE nor BDE can target a verbal projection headed by non-finite perfect *have*, as already demonstrated in §3.3.

Fourth, extraneous conditions must be satisfied to license BDE and PAE when the verb that immediately precedes the ellipsis site displays *-ing* inflection in British English. Based on the examples reported in the literature, it appears that *being* / *doing* can only immediately precede the ellipsis site when the finite verb is a contrastively focused token of *be*; see (52) and (53) (Baker et al. 1989). (Evidence that *-ing*-verbs can appear in a clause that features predicate ellipsis when they do not immediately precede the ellipsis site is provided by the examples in (54), in which ellipsis occurs in a gerundive adjunct clause.)¹⁵

- (52) a. i. * I'm not working, but Bill may be doing. (Pullum & Wilson 1977:761)
ii. * If you're not working, you should be doing. (ibid.)
iii. * Rab is throwing a TV out the window, and Morag is doing, too. (Thoms 2011a:265)
b. i. A: Why do you smile? B: I didn't know I WAS doing. (Halliday & Hasan 1976:122)
ii. A: Why don't you sit quietly? B: I AM doing. (Quirk et al. 1985:875)
iii. Did I talk too much? I felt like I WAS doing. (Algeo 2006:288)
- (53) a. i. * Betsy must have been being hassled by the police, and Peter must have been being, too. (Sag 1976:31)
ii. * Gary was being given a book, and Mary was being, too. (Baker et al. 1989:244)
iii. * Rab is being an idiot, and Mary is being, too. (Thoms 2011a:264)
b. i. A: Why is he being examined by a psychiatrist? B: I didn't know he WAS being. (Pullum & Wilson 1977: fn. 11)
ii. Remember, always be respectful and courteous, even if the officer ISN'T being. (Aelbrecht & Harwood 2015:92)
iii. If Ted wasn't being difficult, then who WAS being? (ibid.: fn. 57)

¹⁴ I put aside here the fact that sentential negation and certain instances of *never* are also licensors of English PAE. These items can license ellipsis independently of T. For instance, if negation is present in a sentence, both standard PAE and BDE can be licensed (*ceteris paribus*) regardless of whether the auxiliary verb in T is contracted; see (i).

(i) John'll arrive late, but Pete {will/'ll} not (do). (Griffiths & Den Dikken 2022: fn. 14).

¹⁵ My consultants judged BDE configurations such as (i), in which perfect *have* is the finite verb, as fully acceptable, which tentatively suggests that the conditions on the presence of *doing* in BD elliptical clauses are less stringent than those on the presence of *being* in PAE. A similar naturally-occurring example was found by Chalcraft 2006; see (13c) (repeated in (ii)).

(i) My car's engine makes strange noises—and has been doing for a while now.

(ii) I'm not working today, but I will be doing tomorrow. (Chalcraft 2006:4)

- (54) a. * Max not having arrived yet and Morgan not {having / doing} either, we decided to wait.
(modified from Aelbrecht 2010:197)
- b. With John having been sacked, and with Pete having been too, the office felt rather empty.
- c. With Pete having been sacked but John {having not (been) / not having been}, the world seemed unfair.
(Griffiths & Den Dikken 2022:123)

In summary, it appears that BDE is merely PAE with the addition of non-finite *do*, contrary to Aelbrecht's and Baltin's claims (for reasons to dismiss Aelbrecht's alleged evidence from gerund clauses for BDE and PAE being distinct forms of ellipsis, see Griffiths & Den Dikken 2022).

Lewis' (2024) analysis of BDE is built on Bošković's (2014) phase-based analysis of English predicate ellipsis. Bošković adopts Harwood & Aelbrecht's (2015) analysis of the English verbal domain, according to which verbal inflectional affixes head their own projections, with verbs head-moving into them (with one exception: *-ing* PF-lowers onto the next lowest verb; see (55), where the 'trace' of PF-lowering is represented by Δ).

- (55) [TP He must₁+T [_{ModP} t₁ [_{InfP} have₂-Ø [_{VP}-PERF t₂ [_{AspectP}-PERF be₃-en [_{VP}-PROG t₃ [_{AspectP}-PROG Δ₄ [_{VoiceP} be-ing₄ [_{VP} hassl₅-ed [_{VP} t₅]]]]]]]]]

Against this backdrop, Lewis proposes that BD is inserted to host an inflectional affix that is stranded by PAE: see (56) for example scenarios.

- (56) a. Helen should write a paper, and Sue should do, too.
 ... [TP Sue should₁+T [ModP t₁ [InfP do-Ø [VoiceP Voice [vP write₂ [vP t₂ a paper]]]]]], too.
- b. Leanne should have written her paper by now, and Bob should have done, too.
 ... [TP Bob should₁+T [ModP t₁ [InfP have_{2-Ø} [vP-PERF t₂ [AspectP-PERF do-ne [VoiceP Voice [vP write₃ [vP t₃ a paper]]]]]]]]]

Lewis' motivation for postulating that BD is inserted to host stray inflection comes from her report that BD cannot co-occur with finite dummy *do* (recall (6a), repeated in (57)). If true, this is captured by her analysis, as, in this configuration, there is no stranded affix for BD to host (58).

- (57) * Tom wrote a paper and Emma did do, too. (Lewis 2024:203, her judgment)
- (58) * ... [TP Bob did+T **do** [VoiceP Voice [vP write₁ [vP t₁ a paper]]]]

As already mentioned in §1, the acceptability judgment associated with this datapoint is unrepresentative, as the co-occurrence of BD with finite dummy *do* is well-attested in English grammars (see Chalcraft 2006 and references therein (59)) and dialect repositories (60). Based on this, it seems implausible that BD is inserted to host stray non-finite verbal inflection.

- (59) a. We don't often go out at weekends, but we *do do* sometimes.
b. Has anything like that ever happened to you? Yes, it *does do* regularly.
c. I don't know if she went to university. I seem to think she *did do*. (Chalcraft 2006:1)
- (60) (repeated from (6))
a. We sat down next to one another and eventually I managed to pluck up the courage to ask him out for a drink with me. He did do—he said yes. www.dialectsarchive.com/england-14
b. So when I was seventeen and they asked me if I could look after the shop, and I did do. www.readingsheffield.co.uk/readers-stories-2/dorothy-norbury

Sailor (2014:55) shows that passive-*be* and passive-*get* are in complementary distribution ((61) and (62)), which strongly suggests that they occupy the same syntactic position, which Sailor identifies as Voice (63).¹⁶

- (61) a. John could {be/get} arrested for wearing that.
b. * John could be gotten arrest(ed) for wearing that.
c. * John could get been arrest(ed) for wearing that.

¹⁶ More accurately, Sailor (2014) identifies the projection that passive *be* / *get* occupies as VoiceInflP, as his theory distinguishes between two voice projections, VoiceInflP and VoiceSynP. For current purposes, this distinction is irrelevant.

- (63) TP > PerfP > ProgP > *VoiceP* (*be/get*) > vP > VP

(64) a. ... [TP The cake [T' will [VoiceP be [_{VP} done [_{VP} eaten t_i]]]]]
 ruled out by Baltin and Aelbrecht as a c-selection failure
 b. ... [TP The cake [T' will [do [_{VoiceP} be [_{VP} [_{VP} eaten t_i]]]]]]]
 ruled out by Baltin and Aelbrecht as the wrong position for do

(65) a. * The cake won't get eaten, but the cookies might get done.
b. The cake won't get eaten, but the cookies might do. (repeated from (46))

- If one maintains that passive-*be* and passive-*get* occupy the same position, and if one maintains that *do* is the realization of *v*, then above-described problem will always arise regardless of what functional projection above *v* one stipulates that *be/get* occupies. Given that there is good reason to believe that passive-*be* and passive-*get* occupy the same position (recall (61) and (62)), the problematic link in this chain of reasoning is the assumption that *do* is the realization of *v*.

I contend that BD is a realization of neither *v* nor Voice. I will demonstrate in the next section that the data discussed in this subsection (and all others) receive a straightforward explanation once BD is treated as occupying a position that dominates the projection housing passive-*be/get*.

It appears that, like PAE, BDE is licensed by T and is compatible with all types of phrasal movement from its ellipsis site (syntactically speaking, at least). In other words, it appears that BDE is merely PAE plus non-finite *do*. However, the informally and experimentally collected evidence presented in §3 suggests that non-finite *do* can be present only when the ellipsis site is headed by a non-raising verb (this is the Raising Verb Averse generalization; see (38)). In this section, I offer a new analysis that captures this syntactic description of BDE.

26

- (67) a. Sally ate dinner.
 b. Sally may eat dinner.
 c. Sally may have been eating dinner.
 d. Sally has been being paid (to eat dinner).
- (68) a. [TP Sally₁ [T' T⁰ [VP t₁ ate_[S, pst] dinner]]]
 b. [TP Sally₁ [T' T⁰+may_[pres] [VP t₁ eat_[S, inf] dinner]]]
 c. [TP Sally₁ [T' T⁰+may_[pres] [VP have_[inf] [VP been_[perf] [VP t₁ eating_[S, prog] dinner]]]]]
 d. [TP Sally₁ [T' T⁰+has_[pres] [VP t₂ [VP been_[perf] [VP being_[prog] [VP paid_[S, pass] t₁]]]]]]]

Furthermore, I assume that the strict order of verbs (i.e. MOD > PERF > PROG > PASS > LEX) has a compositional semantic source: for instance, the function that specifies whether an event is continuous cannot take as its argument a semantic object whose imperfectivity is already specified, which rules out the unattested PROG > PERF word order. This view better aligns with the WYSIWYG syntactic treatment of the English verbal domain adopted here than an analysis that derives the strict order of verbs via syntactic selection, as the latter view requires implausibly complex selectional specifications (e.g., that PERF selects for either PROG or PASS or LEX).

Sailor (2014) and Harwood (2015) marshal abundant evidence from the distribution of VP-topicalization, reduced relative clauses, pseudoclefting, predicate *wh*-questions, particle preposing and other syntactic phenomena to demonstrate that the VP headed by passive *be* / *get* is a 'privileged' phrase (to use Sailor's term) in the English verbal domain, insofar as it behaves more like a thematic verbal projection, such as vP or VP_{LEX}, than an inflectional verbal projection, such as VP_{PERF} or VP_{PROG}. Following Sailor and Harwood, I interpret VP_{PASS}'s privilege as evidence that this verbal projection, when projected, is treated by the syntactic system as belonging to the thematic domain, as opposed to the inflectional domain. Distinguishing between these domains and determining VP_{PASS}'s status within this bipartite system is crucial for specifying British English *do*'s syntactic properties, to which I now turn.

I propose that non-finite British English *do* is a light verbal head (henceforth, DO_[NF]), which heads its own functional projection, *do*P. DO_[NF] is semantically vacuous and optionally projected. DO_[NF] has a syntactic selectional requirement: it can select only for a non-raising thematic domain. This entails that DO_[NF] selects only for the structurally highest V in the thematic domain, and that it can select only for non-raising verbs (i.e. V_[S]). The Vocabulary Insertion rules for DO_[NF] are roughly specified in (69) (cf. Baltin 2012:§5). These rules state that DO_[NF] is realized as /dʌn/ or /duːɪŋ/ when DO_[NF] receives a *perf* or *prog* INFL specification and the entirety of its sister is unpronounced, is realized as /duː/ when DO_[NF] receives either *inf* or no INFL specification and its sister is unpronounced, and is phonologically null in the elsewhere case, that is to say, when any part of its syntactic sister is pronounced.^{17, 18}

- (69) a. /dʌn/ ↔ DO_[NF, perf] / Ø
 b. /duːɪŋ/ ↔ DO_[NF, prog] / Ø
 c. /duː/ ↔ DO_[NF] / Ø
 d. Ø ↔ DO_[NF]

To capture the observation that BDE and PAE display identical licensing conditions and can target for ellipsis the same functional verbal projections, I propose that the ellipsis operation that occurs in BDE and PAE is the same, namely standard PAE. This ellipsis operation, which is licensed syntactically by T, involves the suppression of the visibility of a phrase to the phonological module of grammar (Merchant 2004:670; see Güneş & Lipták 2022 for further discussion and references). I assume that BDE's more restricted distribution is therefore solely due to DO_[NF]'s stringent selectional requirements.

5.1. How this analysis captures the BDE dataset

Restricting DO_[NF] to selecting for non-raising, lexical verbs explains why BD is incompatible with elided copular VPs (recall §3.2), which are headed by the raising verb copula *be*:

¹⁷ If modal / auxiliary verbs syntactically selected for their complements, then the presence of *do*P in the functional sequence would disrupt selection. By adopting the background assumption that modal / auxiliary verbs do not select their sisters, this concern is circumvented.

¹⁸ An anonymous reviewer correctly points out that the Vocabulary Insertion rules in (69) are unconventional insofar as the specified conditioning environment for DO_[NF] is not within the same morphological word as DO_[NF]. I must stress that I view these rules as preliminary and welcome any future attempt to more closely align them with contemporary morphological frameworks. It should be noted that, however these rules are formulated, they must not make direct reference to ellipsis, as *do* may arise whenever its complement is phonologically null, as in VP-preposing, in which *do*'s complement is a trace/copy of movement, rather than an ellipsis site (but see Thoms 2011a for a proposal that unifies ellipsis and copy-deletion).

(i) I expect Rab to win the race, and [win the race]₁ he will **do** t₁.

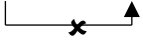
(Thoms 2011a:266)

- (70) a. * Clara might be a teacher, and Sam might do, too.
 b. ... [TP Sam₁ [T' might [doP do [VP be [SC t₁ a teacher]]]]]
 (DO_[NF]'s selectional requirements not met)

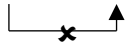
Imposing this selectional restriction on DO_[NF] also explains why BD is incompatible with passive-*be* ellipsis sites but compatible with passive-*get* ones (recall §3.1), why BD is incompatible with raising-*have* ellipsis sites (recall (39)), and why BD is incompatible with progressive-*be* ellipsis sites (recall (45)):

- (71) a. * The cake won't be eaten, but the cookies might do.
 b. ... [TP [the cookies]₁ [T' might [doP do [VP be [VP eaten_[S] t₁]]]]]

- (72) a. The cake won't get eaten, but the cookies might do.
 b. ... [TP [the cookies]₁ [T' might [doP do [VP get_[S] [VP eaten_[S] t₁]]]]]

- (73) a. John hasn't any money left, and Mary hasn't (*done), either.
 b. ... [TP Mary₁ [T' has₂n't [doP done [VP t₁ [V' t₂ [VP t₂ any money left]]]]]]]

- (74) a. # John will be dying to go, and Mary will do, too. (ellipsis site means 'die to go')
 b. ... [TP Mary₁ [T' will [doP do [VP be [t₁ dying_[S] to go t₁]]]]]


Restricting DO_[NF] to selecting for the structurally highest verb in the thematic domain, which is VP_{PASS} when VP_{PASS} is projected, is required to avoid generating the following unattested (and entirely unacceptable) passive-*get* configurations:

- (75) a. * The cake won't get eaten, but the cookies might get done.
 b. ... [TP [the cookies]₁ [T' might [VP get_[S] [doP done [VP eaten_[S] t₁]]]]]

 (get is highest verb in the thematic domain)

Finally, stipulating the Vocabulary Insertion rules in (69) ensures that BD is never overtly present in the absence of ellipsis (76).

- (76) * Sally might eat cake, and Fred might do (*eat cake), too.

Recall from §2.6 that BD can co-occur with pseudogapping. Thoms 2016, which is arguably the most viable recent analysis of English pseudogapping, proposes that pseudogapping involves an overt form of QR (a constrained form of A'-movement) to the Specifier of ΣP, which is immediately dominated by TP:

- (77) Although she won't eat chocolate, [TP she might [ΣP cake₂ [t₁ eat t₂]]].

This analysis of pseudogapping is incompatible with my analysis of BDE, which states that DO_[NF]'s sister is the target for ellipsis. According to my analysis, there is no position into which to leftward-move the pseudogapping remnant that derives the correct word order or creates the correct environment for DO_[NF] to be phonologically expounded (according to (69)), as (78) shows.

- (78) Apparently, the firewoman was able to help John sooner than she could do Paula.
 a. ... she₁ could [ΣP Paula₂ [doP do [t₁ help t₂]]] (incorrect word order)
 b. ... she₁ could [doP do [ΣP Paula₂ [t₁ help t₂]]] (DO_[NF] incorrectly expounded)

To align mine and Thoms' analyses, I tentatively propose that pseudogapping indeed involves overt QR of the remnant, but that this movement targets a rightward adjunction position and is licensed only when string-vacuous (cf. Abels & Dayal 2023 for a similar constraint on overt QR in multiple sluicing) (79). I am confident (but cannot yet be certain) that this change in the movement's landing-site is compatible with each of the datapoints and arguments Thoms marshals for his analysis.

(79) ... she₁ could [_{ΣP} [_{doP} do [*t*₁ help *t*₂]] Paula₂]

To summarize, my syntactic analysis captures the unique properties of BDE by treating British English *do* as a functional head that selects only for the structurally highest verb in the thematic domain. Note that this analysis makes no attempt to encode locality conditions on A'-movement from anywhere within the BD elliptical clause, as the experimental results reported in §2 strongly suggest that no such locality conditions obtain.

6. Loose ends

6.1. The source of the amelioration effect of *have done*

Recall from §2 that, based on suggestions from my 5 consultants, four of my eight experiments included AUX—that is to say, whether the introducer verb is perfect *have* or a modal verb—as a factor. Recall also that the presence of perfect *have* (as opposed to a modal) had an ameliorating effect on the acceptability of only BDE in only two of these experiments. This amelioration effect was statistically significant in BDE configurations that select for a copular phrase ellipsis site and in BDE configurations involving ACD in comparative clauses, and statistically insignificant in BDE configurations involving ACD in *that*-relatives.

Two questions arise: (i) why should using perfect *have done* (as opposed to e.g., *will do*) provide an amelioration effect, and (ii) why should this amelioration effect arise in some BDE configurations and not others?

Regarding question-(i), appealing to frequency would be wrong-footed, as Oger (2019) shows that, in the spoken part of British National Corpus, *do* comprises 64% of all tokens of BD, followed by *done* (36%) and then *doing* (<1%). This finding therefore discredits Huddleston & Pullum's (2002:1525) impressionist claim that the past participle form of BD is the most frequently used form. Another possibility is that the amelioration effect has a syntactic source. For instance, one might claim that phrases of differing sizes and/or syntactic statuses are elided in '*have done*' and '*modal + do*' BDE configurations, and that eliding one of these phrases is preferred to eliding the other, for syntactic reasons (e.g., one is a full phrase, the other is a phase head's complement; see Bošković 2014 and Lewis 2024 for analyses of PAE and BDE that appeal to eliding different parts of phrases). This possibility seems unlikely however, as (i) the size of the amelioration effect is small (often too small to yield statistical significance), which is uncharacteristic of a syntactic violation, and (ii) when the amelioration is observed, BDE does **not** target phrases of different sizes / statuses, if the analysis from §5 is correct.

In §2, I suggested that the ameliorating effect of using *have done* has a prosodic source. The motivation behind this suggestion comes from the fact that BD is well-known to be an enclitic (Haddican 2007) and therefore sensitive to its prosodic environment (see Thoms & Sailor 2018 for an analysis that exploits BD's enclitic status). Exactly how *have done* provides a prosodic amelioration effect (assuming it does) remains to be determined, however. The obvious first possibility to explore is whether the difference in vowel length between *do* (/du:/) and *done* (/dʌn/ or /dʊn/), or the fact that only *done* contains a coda consonant, yields a difference in prosodic (extrametrical) foot formation (cf. Hayes 1982) in the elliptical clause, such that prosodic feet are more simply / regularly formed with *done* than *do*, which is reflected in acceptability judgments. I leave exploring this possibility, and whether it can be developed into a prosodic analysis that has sufficient explanatory power to predict in which BDE configurations an amelioration effect will be observed, for future research.

6.2. Acceptability judgments on BDE that still require experimental confirmation

As mentioned already, several informal acceptability judgments on BDE still require experimental confirmation. These include the judgments attributed to the BDE examples presented in §3, and also claims from the literature about the (im)possibility of covert extraction from the BD ellipsis site (see Baltin 2007, 2012, Thoms & Sailor 2018, Lewis 2024). The latter cases were excluded from the experiment series reported in §2 because testing these claims requires adopting a different task-type to the one employed in my experiment series (a forced-choice task, as opposed to a Likert scale task).

6.3. 'British' *do* in *as*- and *which*-parentheticals in American English varieties

As mentioned in footnote 2, non-finite, 'propredicate' *do* is attested in *as*- and *which*-parenthetical clauses in both British and American English (the examples below are from Kato & Butters 1987):

- (80) a. That was when the goons, as he called them, were expected to seize the ballot boxes, as they allegedly had done in the 1984 legislative election. (*Time*, 17 February 1986:8)
 b. He did not take his medicine, *which* he should have done.

Further research is required to determine whether there are multiple forms of proredicate *do* across English (some dialect-specific, some not) or whether propredicate *do* is exceptionally licensed only in *as*- and *which*-parenthetical clauses in American English because of some special syntactic property that these clauses display.

7. Conclusion

In this article I reported the results of eight acceptability judgment experiments on British English *do*-ellipsis (BDE). These results revealed that the BDE site is permeable to all forms of overt A'-movement, contrary to claims from the literature, which are based on informally-collected data. These results also revealed that British *do* cannot immediately precede an ellipsis site headed by a raising verb (i.e., a verb that can raise to T), an observation I suggested should be encoded in *do*'s selectional requirements. Assuming that these results are reliable, they automatically invalidate previous analyses of BDE that treat ellipsis-licensed phrases as syntactically opaque domains (Aelbrecht 2010, Baltin 2012, Lewis 2024), and they therefore function to cast suspicion on the prevalent claim that elided phrases are 'spelled-out' domains, equal / akin to phases (Takahashi 2004, Gengel 2007, Gallego 2009, Aelbrecht 2010, Rouveret 2012, Bošković 2014, Sailor 2018, Manetta 2020, among others).

Funding. This study was funded by the University of Tübingen and the German Research Foundation (Deutsche Forschungsgemeinschaft), project number 522896622.

Financial interests. The author declares that he has no financial interests.

Acknowledgments. I thank the students who attended my Wintersemester 2023-24 seminar *Data Collection in Linguistics* for helping to develop the experimental stimuli, and I thank Annabell Bracun and Magdalena Horáčková for their proof-reading efforts. I also thank the audiences of GLOW 47, CGG 33, EACBE 6, and the Potsdam Morpho-Syntax Lab, three anonymous reviewers, and the editors Daniel Harbour and Martin Salzmann, whose insightful comments and valuable feedback helped to significantly improve this article.

References

- Abels, Klaus. 2012. *Phases: An essay on cyclicity in syntax*. Berlin: De Gruyter. <https://doi.org/10.1515/9783110284225>
- Abels, Klaus and Venetta Dayal. 2023. On the Syntax of Multiple Sluicing and What It Tells Us about Wh-Scope Taking. *Linguistic Inquiry* 54(3): 429–477. https://doi.org/10.1162/ling_a_00448
- Aelbrecht, Lobke. 2010. *The syntactic licensing of ellipsis*. Amsterdam: John Benjamins. <https://doi.org/10.1017/s0022226710000423>
- Aelbrecht; Lobke and William Harwood. 2015. To be or not to be elided: VP ellipsis revisited. *Lingua* 153: 66–97. <https://doi.org/10.1016/j.lingua.2014.10.006>
- Algeo, John. 2006. *British or American English? A Handbook of Word and Grammar Patterns*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511607240>
- Baker, C. Lee. 1984. Two observations on British English *do*. *Linguistic Inquiry* 15: 155–157.
- Baker, Mark, Kyle Johnson, and Ian Roberts. 1989. Passive Arguments Raised. *Linguistic Inquiry* 20(2): 219–251.
- Baltin, Mark. 2006. The Nonunity of VP-Preposing. *Language* 82(4): 734–766. <https://doi.org/10.1353/lan.2006.0181>
- Baltin, Mark. 2007. The position of adverbials. In *Phrasal and Clausal Architecture : Syntactic Derivation and Interpretation: in Honor of Joseph E. Emonds*, eds. Simin Karimi, Vida Samiian, and Wendy K. Wilkins, 25–39. Amsterdam: John Benjamins. <https://doi.org/10.1075/la.101>
- Baltin, Mark. 2012. Deletion versus proforms: An overly simple dichotomy? *Natural Language and Linguistic Theory* 30: 381–423. <https://doi.org/10.1007/s11049-011-9157-x>
- Bošković, Željko. 2014. Now I'm a Phase, Now I'm Not a Phase: On the Variability of Phases with Extraction and Ellipsis. *Linguistic Inquiry* 45(1): 27–89. https://doi.org/10.1162/LING_a_00148
- Chalcraft, Faye. 2006. *Do doubling in varieties of English*. Paper presented at the Workshop on Syntactic Doubling, Amsterdam, March 2006. https://www.meertens.knaw.nl/projecten/edisynt/Online_proceedings/Paper_Chalcraft.pdf. Accessed 03.07.2025
- Chomsky, Noam. 1977. On wh-movement. In *Formal syntax*, ed. Peter Culicover, Thomas Wasow, and Adrian Akmajian, 71–132. New York: Academic Press.
- Chomsky, Noam. 2000. Minimalist inquiries: The framework. In *Step by Step: Essays on Minimalist Syntax in Honor of Howard Lasnik*, eds. Roger Martin, David Michaels, and Juan Uriagereka, 89–155. Cambridge, MA: MIT Press.

- Chomsky, Noam. 2001. Derivation by phase. In *Ken Hale. A Life in Language*, ed. Michael Kenstowicz, 1–52. Cambridge, MA: MIT Press. <https://doi.org/10.7551/mitpress/4056.001.0001>
- Christensen, Rune. 2023. ordinal—Regression Models for Ordinal Data. R package version 2023.12-4.1. <https://CRAN.R-project.org/package=ordinal>
- Chung, Sandra, William Ladusaw & James McCloskey. 1995. Sluicing and logical form. *Natural Language Semantics* 3: 239–282. <https://doi.org/10.1007/BF01248819>
- Cortés Rodríguez, Álvaro. 2022. Multiple sluicing and islands: a cross-linguistic experimental investigation of the clausemate condition. *The Linguistic Review* 39(3): 425–455. <https://doi.org/10.1515/tlr-2022-2093>
- Di Paolo, Marianna. 1993. Propredicate *do* in the English of the Intermountain West. *American Speech* 68(4): 339–356. <https://doi.org/10.2307/455771>
- Featherston, Sam. 2020. Can we build a grammar on the basis of judgements? In *Linguistic Intuitions, Evidence and Method*, eds. Samuel Schindler, Anna Drożdżowicz, and Karen Brøcker, 165–188. Oxford: Oxford University Press. <https://doi.org/10.1093/oso/9780198840558.001.0001>
- Fox, Danny. 2002. Antecedent-contained Deletion and the Copy Theory of Movement. *Linguistic Inquiry* 33(1): 63–96. <https://doi.org/10.1162/002438902317382189>
- Gallego, Ángel J. 2009. Ellipsis by phase. Paper presented at the 19th Colloquium on Generative Grammar. https://www.academia.edu/download/30814226/GAL_vitoria.pdf. Accessed 03.07.2025
- Gerbrich, Hannah, Vivian Schreier, and Sam Featherston. 2019. Standard items for English judgement studies: Syntax and semantics. In *Experiments in focus: Information structure and semantic processing*, eds. Sam Featherston, Robin Hörnig, Sophie von Wietersheim, and Susanne Winkler, 305–328. Berlin: De Gruyter. <https://doi.org/10.1515/9783110623093>
- Gengel, Kirsten. 2007. *Pseudogapping and ellipsis*. Oxford: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199665303.001.0001>
- Griffiths, James. 2019. Beyond MaxElide: An Investigation of $\bar{A}$ -Movement from Elided Phrases. *Linguistic Inquiry* 50(3): 571–607. https://doi.org/10.1162/ling_a_00317
- Griffiths, James and Marcel den Dikken. 2022. English VP ellipsis in Unusual Subject configurations: Reviving the Spec-Head agreement approach. In *The derivational timing of ellipsis*, eds. Güliz Güneş and Anikó Lipták, 97–130. Oxford: Oxford University Press. <https://doi.org/10.1093/oso/9780198849490.003.0003>
- Güneş, Güliz and Anikó Lipták (eds.). 2022. *The derivational timing of ellipsis*. Oxford: Oxford University Press. <https://doi.org/10.1093/oso/9780198849490.001.0001>
- Haddican, William. 2007. The structural deficiency of verbal proforms. *Linguistic Inquiry* 38(3): 539–547. <https://doi.org/10.1162/ling.2007.38.3.539>
- Halliday, Michael A. K and Ruqaiya Hasan. 1976. *Cohesion in English*. London: Longman. <https://doi.org/10.4324/9781315836010>
- Harwood, William. 2015. Being progressive is just a phase: celebrating the uniqueness of progressive aspect under a phase-based analysis. *Natural Language & Linguistic Theory* 33: 523–573. <https://doi.org/10.1007/s11049-014-9267-3>
- Hayes, Bruce. 1982. Extrametricality and English Stress. *Linguistic Inquiry* 13(2): 227–276.
- Huddleston, Rodney and Geoffrey K. Pullum. 2002. *The Cambridge Grammar of the English Language*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/9781316423530>
- Joos, Martin. 1964. *The English verb: Form and meaning*. Madison, WI: University of Wisconsin Press.
- Kato, Kazuo and Ronald R. Butters. 1987. American instances of propredicate *do*. *Journal of English Linguistics* 20: 216–216. <https://doi.org/10.1177/007542428702000205>
- Kennedy, Christopher and Jason Merchant. 2000. Attributive comparative deletion. *Natural Language and Linguistic Theory* 18: 89–146. <https://doi.org/10.1023/a:1006362716348>
- King, Harold. 1970. On blocking the rules for contraction in English. *Linguistic Inquiry* 1: 134–36.
- Lechner, Winfried. 2004. *Ellipsis in Comparatives*. New York: Mouton de Gruyter.
- Lewis, Rebecca. 2024. British English *do*-ellipsis is full phase ellipsis. In *Proceedings of the 40th West Coast Conference on Formal Linguistics*, ed. Jiayi Lu, Erika Petersen, Anissa Zaitsu, and Boris Harizanov, 202–212. Somerville, MA: Cascadilla Proceedings Project. <https://www.lingref.com/cpp/wccfl/40/paper3712.pdf>
- Lipták, Anikó and Jason Merchant. 2019. Extraction from copulas is bled by ellipsis: On the interaction of head movement, phrasal movement and ellipsis. Paper presented at *Sluicing and ellipsis at 50*. University of Chicago, April 2019.
- Manetta, Emily. 2020. Expanding the typology of Verb Second VPE: The case of Kashmiri. In *Rethinking verb second*, eds. Teresa Biberauer, Sam Wolfe, and Rebecca Woods, 723–744. Oxford: Oxford University Press. <https://doi.org/10.1093/oso/9780198844303.003.0031>
- Merchant, Jason. 2004. Fragments and ellipsis. *Linguistics and Philosophy* 27.6:661–738.
- Merchant, Jason. 2008a. An asymmetry in voice mismatches in VP-ellipsis and pseudogapping. *Linguistic Inquiry* 39(1): 169–179. <https://doi.org/10.1162/ling.2008.39.1.169>

- Merchant, Jason. 2008b. Variable island repair under ellipsis. In *Topics in Ellipsis*, ed. Kyle Johnson, 132–153. Cambridge: Cambridge University Press. <https://doi.org/10.1017/cbo9780511487033.006>
- Messick, Troy and Gary Thoms. 2016. Ellipsis, Economy, and the (Non)uniformity of Traces. *Linguistic Inquiry* 47(2): 306–332. https://doi.org/10.1162/ling_a_00212
- Miller, Phillip and Geoffrey Pullum. 2013. Exophoric VP Ellipsis. In *The Core and the Periphery: Data-Driven Perspectives on Syntax Inspired by Ivan A. Sag*, eds. Philip Hofmeister and Elisabeth Norcliffe, 1–28. Chicago: CSLI Publications.
- Oger, Kimberly. 2019. A Study of Non-Finite Forms of Anaphoric do in the Spoken BNC. *Anglophonia. French Journal of English Linguistics*, (28). <https://doi.org/10.4000/anglophonia.2936>
- Park, Myung-Kwan. 2013. British English do and extraction out of vP. *Linguistic Research* 30(1): 1–17. <https://doi.org/10.17250/khisli.30.1.201304.001>
- Peirce, Jonathan, Jeremy Gray, Sol Simpson, Michael MacAskill, Richard Höchenberger, Hiroyuki Sogo, Erik Kastman, and Jonas Lindeløv. 2019. PsychoPy2: experiments in behaviour made easy. *Behavior Research Methods* 51: 195–203. <https://doi.org/10.3758/s13428-018-01193-y>
- Pullum, Geoffrey. & Deirdre Wilson. 1977. Autonomous syntax and the analysis of auxiliaries. *Language* 53(4): 741–788. <https://doi.org/10.2307/412911>
- Quirk, Randolph, Sidney Greenbaum, Geoffrey Leech, and Jan Svartvik. 1985. *A Comprehensive Grammar of the English Language*. London: Longman.
- Rouveret, Alain. 2012. VP ellipsis, phases, and the syntax of morphology. *Natural Language and Linguistic Theory* 30: 897–963. <https://doi.org/10.1007/s11049-011-9151-3>
- Sag, Ivan. 1976. Deletion and Logical Form. Doctoral dissertation, Massachusetts Institute of Technology. <https://dspace.mit.edu/bitstream/handle/1721.1/16401/03491413.pdf?sequence=1>. Accessed 03.07.2025
- Sailor, Craig. 2014. The variables of VP ellipsis. Doctoral dissertation, University of California, Los Angeles. https://linguistics.ucla.edu/wp-content/uploads/2021/11/CraigSailor_dissertation.pdf. Accessed 03.07.2025
- Sailor, Craig. 2018. The typology of head Movement and ellipsis: A reply to Lipták & Saab. *Natural Language & Linguistic Theory* 36: 851–875. <https://doi.org/10.1007/s11049-017-9391-y>
- Schiele, Miriam L. 2025. Contrastivity in left branch extraction in German: An experimental approach. Paper presented at ConSOLE 33, University of Göttingen, Jan 2025.
- Schwarzer, Luise. 2023. Subject co-reference in antecedent-contained deletion. In *Proceedings of the 53rd meeting of the North East Linguistics Society*, eds. Suet-Ying Lam and Satoru Ozaki. GLSA, University of Massachusetts.
- Sprouse, Jon and Diogo Almeida. 2012. Assessing the reliability of textbook data in syntax: Adger's Core Syntax. *Journal of Linguistics* 48: 609–652. <https://doi.org/10.1017/s0022226712000011>
- Sprouse, Jon and Diogo Almeida. 2017. Design sensitivity and statistical power in acceptability judgment experiments. *Glossa* 2(1): 14, 1–32. <https://doi.org/10.5334/gjgl.236>
- Sprouse, Jon and Diogo Almeida. 2018. Setting the empirical record straight: Acceptability judgments appear to be reliable, robust and replicable. *Behavioral and Brain Sciences* 40. <https://doi.org/10.1017/s0140525x17000590>
- Sprouse, Jon, Carson Schütze and Diogo Almeida. 2013. A comparison of informal and formal acceptability judgments using a random sample from *Linguistic Inquiry* 2001–2010. *Lingua* 134: 219–248. <https://doi.org/10.1016/j.lingua.2013.07.002>
- Stockwell, Richard. 2020. Contrast and Verb Phrase Ellipsis: Triviality, Symmetry, and Competition. Doctoral dissertation, University of California Los Angeles. <https://escholarship.org/uc/item/2x4843h7>. Accessed 03.07.2025
- Takahashi, Shoichi. 2004. Pseudogapping and cyclic linearization. In *Proceedings of the 43rd Meeting of the North East Linguistic Society*, eds. Matthew Wolf and Keir Moulton, 571–85. Amherst, MA: GLSA.
- Takahashi, Shoichi and Danny Fox. 2005. MaxElide and the Re-binding Problem. In *Proceedings of Semantics and Linguistic Theory* 15, ed. Efthymia Georgala and Jonathan Howell, 223–240.
- Thoms, Gary. 2011a. 'Verb floating' and VP-ellipsis: Towards a movement account of ellipsis licensing. *Linguistic Variation Yearbook* 10: 255–300. <https://doi.org/10.1075/livy.10.07tho>
- Thoms, Gary. 2011b. Review of Lobke Aelbrecht's *The syntactic licensing of ellipsis*, *Journal of Linguistics* 47(1): 219–225. <https://doi.org/10.1017/S0022226710000423>
- Thoms, Gary. 2016. Pseudogapping, parallelism and the scope of focus. *Syntax* 19(3): 286–307. <https://doi.org/10.1111/synt.12122>
- Thoms, Gary. 2019. Varieties of English. In *The Oxford Handbook of Ellipsis*, eds. Jeroen van Craenenbroeck and Tanja Temmerman, 1020–39. Oxford: Oxford University Press.
- Thoms, Gary and Craig Sailor. 2018. When silence gets in the way: Extraction from do-ellipsis in British dialects. *Proceedings of the 48th meeting of the North East Linguistics Society*, 145–155.

- Thompson, Andrea. 2014. Beyond deep and surface – Explorations in the typology of anaphora. Doctoral dissertation, University of California Santa Cruz. escholarship.org/uc/item/1ch3f905. Accessed 03.07.2025
- Warner, Anthony. 1985. *The Structure of English Auxiliaries: A Phrase Structure Grammar*. Bloomington: Indiana University Linguistics Club.